Bearing Capacity For Foundation Design

Rev.	Rev. Date	Description	Prepared By	Checked By	Approved By	Employer
A01	31 Jan 2019	For Information	J.H.JEON	M.G.LEE	J.J.YANG	

1. INTRODUCTION 3

1.1. PURPOSE 3

1.2. CODES, STANDARDS AND PROGRAMS 3

1.3. SITE ELEVATION 3

2. SOIL PARAMETERS 5

2.1. BOREHOLE LOCATION MAP. 5

2.2. APPLIED SOIL PARAMETERS 7

3. PRINCIPLE OF FOUNDATION DESIGN 8

3.1. ANALYSIS AND DESIGN OF SHALLOW FOUNDATION 8

3.1.1. Soil Bearing Capacity Calculation 8

3.1.2. Settlement Calculation 10

3.1.3. Allowable Settlement Criteria 12

3.2. ANALYSIS AND DESIGN OF PILES 13

3.2.1. Vertical Capacity of Pile 13

3.2.2. Lateral Capacity of Pile 18

4. SUMMARY OF SHALLOW FOUNDATION ANALYSIS RESULT 20

4.1 BORE HOLE LOCATION FOR SHALLOW FOUNDATION MAP 20

4.2 SHALLOW FOUNDATION ANALYSIS RESULT TABLE 22

5. SUMMARY OF PILE FOUNDATION ANALYSIS RESULT 24

5.1 BORE HOLE LOCATION FOR PILE FOUNDATION MAP 24

5.2 PILE CAPACITY ANALYSIS RESULT TABLE 26

APPENDIX 28

I) SAMPLE SHALLOW FOUNDATION CALCULATION SHEET 28

II) SAMPLE VERTICAL PILE CAPACITY CALCULATION SHEET 43

III) SAMPLE LATERAL PILE CAPACITY CALCULATION SHEET 56

IV) SAMPLE PILE STRUCTURAL CAPACITY CHECK CALCULATION SHEET 67

ATTACHMENT-1: BORING HOLE LOG 81

ATTACHMENT-2: LABORATORY TEST RESULT 81

ATTACHMENT-3 : PILE CAPACITY ANALYSIS RESULTS FOR ALL BH LOGS EXCEPT BH ON FUTURE AREA 81

1. INTRODUCTION

1.1. Purpose

The purpose of this document is to determine the bearing capacity of soil for the design of shallow and pile foundations to be applied for the onshore structures of Ma'aden Ammonia III project at ras al kahir, Saudi Arabia.

1.2. Codes, Standards and Programs

The design, detailing, material specification, testing, etc. shall be conformed in accordance with employer's requirement which may include the following Standards, Codes, manuals and programs.

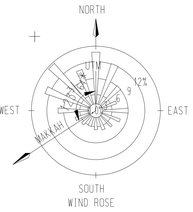
- Ma'aden Engineering Manual
- AASHTO LRFD Brdige Design specifications 7th Edition
- ALL-PILE: Software Program used for the analysis of lateral Capacity of Pile

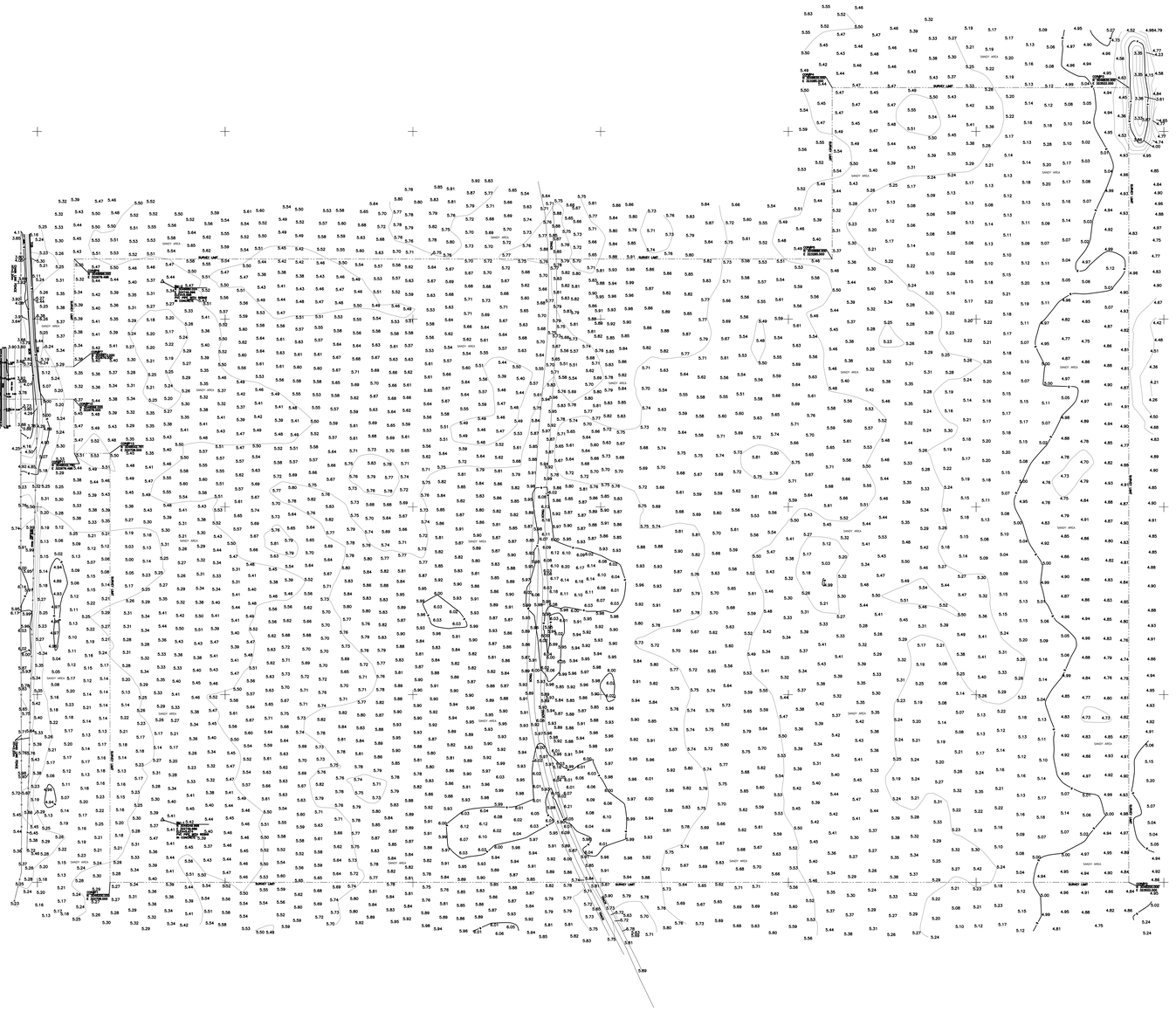
1.3. Site Elevation

As per the recent site investigation and topographic survey study, project site's natural ground elation (N.G.L.) varies from M.S.L. 5.3~6.01m. The proposed final ground elevation shall be M.S.L+7.00m (E.L.+0.00).
The below figure shows the sites topographic survey result.

E 322,350m | E 322,500m | E 322,650m | E 322,800m | E 322,950m | E 323,100m | E 323,250m | E 323,400m | E 323,550m

N 3,046,950m | N 3,046,800m | N 3,046,650m | N 3,046,500m | N 3,046,350m | N 3,046,200m | N 3,046,050m





SCALE 1:1500

SURVEY NOTES

1) COORDINATES ARE UTM (ZONE 39) INTERNATIONAL 1924 ELLIPSOID. VALUES IN METER. HORIZONTAL DATUM: 1994 HORIZONTAL ADJUSTMENT (AIN AL'ABD 1970)
2) VERTICAL DATUM: MEAN SEA LEVEL (M.S.L) 1978 SAUDI ARAMCO VERTICAL DATUM (SAVD 78) 1994 ADJUSTMENT.
3) DATE OF SURVEY: DEC. 2018
4) ALL TOPOGRAPHIC FEATURES WERE SURVEYED USING CONVENTIONAL /GPS SURVEYING EQUIPMENT AND METHODS WITH PROBABLE HORIZONTAL & VERTICAL ACCURACY OF 0.05/0.1m RESPECTIVELY.
5) CONTOURS CREATED BY TRIANGULATED IRREGULAR NETWORK (TIN) AS PER SOW.
6) CONTOURS INTERVAL IS 0.25m FOR MINOR WHILE 1m FOR MAJOR CONTOUR.
7) REFERENCE MAADEN SURVEY MONUMENT USED FOR THIS PROJECT ARE. BM-MIC-1, E=320910.117, N=3042895.651, ELEV=3.505 & MIC-3 E=319279.273, N=3046292.043, ELEV=5.222
8) COMBINED SCALE FACTOR= 0.999993400

LEGEND

△	CONTROL STATION
	LIGHT POST
□○ MH	MANHOLE
	SIGN BOARD
	ROAD SIGN
	VENT PIPE
	GATE
	FENCE
	SURVEY LIMIT
4.45	SPOT ELEVATION
4.60	MANHOLE RIM ELEVATION
	MAJOR CONTOUR
	MINOR CONTOUR

ABBREVIATIONS:

TOB	TOP OF BANK/SLOPE	BOB	BOTTOM OF BANK/SLOPE
HT	HEIGHT	MHE	MANHOLE ELECTRIC

CONTROL POINTS LIST

POINT	EASTING	NORTHING	ELEV.
BM-1	322749.986	3046249.998	5.674
BM-2	322749.996	3046680.024	5.698
MIC-1	320910.117	3042895.651	3.505
MIC-3	319279.273	3046292.043	5.222

DOCUMENT APPRAISAL STAMP

MA'ADEN Saudi Arabian Mining Company | JACOBS

REVIEWER'S NAME	SIGN	DATE

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Result	Code	Description
☐	A	Accepted
☐	C	Proceed with exceptions to next submission and Revision
☐	R	Do Not Proceed – Revise as noted and Resubmit

Acceptance in any of these categories in no way relieves the Contractor/Supplier of its responsibility for the due & proper performance of the Works in accordance with the Contract/Purchase Order

PROJECT MANAGER SIGN-OFF:

NAME | DATE:

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-522-7000-EG-CV-TPD-0002	TOPOGRAPHIC & CONTOUR MAP (1)
MD-522-7000-EG-CV-TPD-0003	TOPOGRAPHIC & CONTOUR MAP (2)
MD-522-7000-EG-CV-TPD-0004	TOPOGRAPHIC & CONTOUR MAP (3)
MD-522-7000-EG-CV-TPD-0005	TOPOGRAPHIC & CONTOUR MAP (4)

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		C01	23.JAN.'19	ISSUED FOR CONSTRUCTION	J.H. JEON	S.W. CHOI	M.G. LEE	J.J. YANG	
		A01	09.JAN.'19	ISSUED FOR REVIEW	J.H. JEON	S.W. CHOI	M.G. LEE	J.J. YANG	

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APPROVED	J.J. YANG	23-JAN-2019
EMPLOYER		
SCALE 1:1500	A0	

MA'ADEN AMMONIA III PROJECT
TOPOGRAPHIC & CONTOUR OVERALL MAP

DRAWING NUMBER MD-522-7000-EG-CV-TPD-0001 REV C01

2. SOIL PARAMETERS

2.1. Borehole Location Map.

The distribution of boreholes across the project site can be seen in the following figure.

DRAWING NUMBER –
REV P06

NOTE

1. ALL COORDINATES ARE IN METERS.
2. PLEASE REFER TO GEOTECHNICAL INVESTIGATION PLAN_P02 FOR TR-07, TR-08, TR-09, TR-10

NORTH
WEST
EAST
SOUTH
WIND ROSE
TR-10

BATTERY LIMIT (UTM N. 3,046,698.000)

BATTERY LIMIT E. 322,708.000

BATTERY LIMIT E. 323,488.030

BATTERY LIMIT N. 3,046,200.000

OF ROAD (UTM E. 322,731.000)

OF ROAD (UTM E. 323,027.500)

OF ROAD (UTM E. 323,154.000)

OF ROAD (UTM E. 323,294.000)

OF ROAD (UTM E. 323,465.000)

OF ROAD (UTM N. 3,046,598.000)

OF ROAD (UTM N. 3,046,484.000)

OF ROAD (UTM N. 3,046,443.000)

OF ROAD (UTM N. 3,046,298.000)

OF ROAD N. 3,046,338.000

OF ROAD N. 3,046,308.000

OF ROAD N. 3,046,211.000

FUTURE TANKAGE AREA

MAIN GATE

NEW FIRE WATER RETENTION POND 90m x 90m x 4.5m DEEP

NEW IRRIGATION POND 45m x 45m x 4.5m DEEP

BH-93 BH-151 BH-94 BH-95 BH-96 BH-97 BH-98 BH-99 BH-100 BH-101 BH-102 BH-103 BH-104 BH-105 BH-106 BH-107 BH-108 BH-109 BH-110 BH-111 BH-112 BH-113 BH-114 BH-115 BH-116 BH-117 BH-118 BH-119 BH-120 BH-121 BH-122 BH-123 BH-124 BH-125 BH-126 BH-127 BH-128 BH-129 BH-130 BH-131 BH-132 BH-133 BH-134 BH-135 BH-136 BH-137 BH-138 BH-139 BH-140 BH-141 BH-142 BH-143 BH-144 BH-145 BH-01 BH-02 BH-03 BH-04 BH-05 BH-06 BH-07 BH-08 BH-09 BH-10 BH-11 BH-12 BH-13 BH-14 BH-15 BH-16 BH-17 BH-18 BH-19 BH-20 BH-21 BH-22 BH-23 BH-24 BH-25 BH-26 BH-27 BH-28 BH-29 BH-30 BH-31 BH-32 BH-33 BH-34 BH-35 BH-36 BH-37 BH-38 BH-39 BH-40 BH-41 BH-42 BH-43 BH-44 BH-45 BH-46 BH-47 BH-48 BH-49 BH-50 BH-51 BH-52 BH-53 BH-54 BH-55 BH-56 BH-57 BH-58 BH-59 BH-60 BH-61 BH-62 BH-63 BH-64 BH-65 BH-66 BH-67 BH-68 BH-69 BH-70 BH-71 BH-72 BH-73 BH-74 BH-75 BH-76 BH-77 BH-78 BH-79 BH-80 BH-81 BH-82 BH-83 BH-84 BH-85 BH-86 BH-87 BH-146 BH-147 BH-148 BH-149 BH-150 BH-152 BH-153

CPT-01 CPT-02 CPT-03 CPT-04 CPT-05 CPT-06 CPT-07 CPT-08

ER-01 ER-02 ER-03 ER-04 ER-05 ER-06 ER-07 ER-08 ER-09 ER-10 ER-11 ER-12 ER-13 ER-14 ER-15 ER-16 ER-17 ER-18 ER-19

TR-01 TR-02 TR-03 TR-04 TR-05 TR-06

CHT-01 CHT-02 CHT-03 CHT-04

DHT-01 DHT-02 DHT-03 DHT-04 DHT-05 DHT-06 DHT-07 DHT-08

LEGEND

- : BOREHOLES LOCATION (BH) - 148 NOS
- : CROSSHOLE TEST (CHT) - 4 NOS
- : DOWNHOLES TEST (DHT) - 8 NOS
- : CONE PENETRATION TEST (CPT) - 8 NOS
- : ELECTRICAL RESISTIVITY TEST (ER) - 19 NOS
- : THERMAL CONDUCTIVITY TEST (TR) - 10 NOS

METERS
0 20 40 60 80 100 120

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				–	–	–	–	–	–	–	–

$FILE$ $TIME$ $DATE$

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DAELIM JACOBS MA'ADEN معادن Saudi Arabian Mining Company

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CHECKED	–	–	
ENGINEER	–	–	
APPROVED	–	–	
EMPLOYER			
SCALE	1:1000	A0	

MA'ADEN AMMONIA PROJECT
ATTACHMENT 5
GEOTECHNICAL INVESTIGATION PLAN (1/2)

DRAWING NUMBER –
REV P06

2.2. Applied Soil Parameters

Soil parameters for the design of foundation are applied by the same principle that was used for Ammonia-II project in which soil parameters are selected based on the range of N-value as agreed upon in Ammonia-II Project. Below is the captured MOM of Ammonia-II Soil parameter determination agreement.

MD-513-M-CV-0001

MA'ADEN معادن FLUOR	Umm Wu'll Phosphate Project Ammonia Plant	DAELIM

Minutes of Meeting

MEETING SUBJECT : Geotechnical Investigation Report Consulting

ATTENDEES	POSITION	COMPANY	
Mr. Abdullah S. Al. Shahrani	Lead Civil Engineer.	Maaden	
Dr. Saad Ali Aiban	Consultant	MAEEN-eng	
Ms. Min-Gyu, Lee	Civil LE	Daelim	**MEETING DATE:** Feb. 10~12, 2014
Mr. Sung-Jun, Jung	Geotechnical Engineer	Daelim	**MEETING TIME:** 10:00 AM ~ 06:00 PM
Mr. Chung-Gu, Youn	Geotechnical Engineer	Daelim	
Mr. Kwang-Deok, Ahn	Civil Engineer	Daelim	

Item No.	DISCUSSION ITEMS	Action By:	Agreed Completion Date	Actual Date Completed
1.0	Factor of Safety - Factor of safety of 2 for piles in compression is the minimum to be used as per all international and local codes and standards. However, factor of safety of 2 for piles in compression (both side and tip resistance) is adopted in accordace with Annex A part VIII Attachment 4 civil, structural and building section 3.9.	Daelim	Completed	Completed
2.0	Angle of internal friction - The angle of internal friction for sands is adopted to have a maximum value of 39 degrees for SPT N value of more than 50.	Daelim	Completed	Completed
3.0	The Modulus of Elasticity (E_s) Values are adpoted as follows : - For soil layer of SPT N value of 5 and smaller, E_s = 300 (N+6) and the E_s shall not exceed 3MPa. - For other soil layer, E_s = 1000 N and the E_s shall not exceed 50MPa. - Sabhka is soil layer of SPT N value 5 and smaller.	Daelim	Completed	Completed
4.0	Settlement for shallow foudnation - Tolerable settlement depends on equipment type and foundation type (spread, combined and mat).	Daelim	Completed	Completed
5.0	Test Piles - A static axial compression test pile is to be added to confirm the pile capacity of East Ammonia storage tank (31210-TNK-002) - A lateral test pile is to be changed to static axial compression test pile (TP-6). - Test piles for tank area are revised and changed as attachment 4.	Daelim	Completed	Completed
6.0	Pile and shallow foundation design parameter - Pile capacity and settlement of foundation are to be designed by using the given parameters as above and in accordance with applicable specifications and standards.	Daelim	Completed	Completed
7.0	Construction - Construction need to be done in a way to assure that the above given parameters are achieved.	Daelim	Completed	Completed

3. PRINCIPLE OF FOUNDATION DESIGN

3.1. Analysis and Design of Shallow Foundation

3.1.1. Soil Bearing Capacity Calculation

I. <u>Terzaghi's bearing capacity theory:</u>

Based on Terzaghi's bearing capacity theory, column load P is resisted by shear stresses at edges of three zones under the footing and the overburden pressure, q (=γD) above the footing. The first term in the equation is related to cohesion of the soil. The second term is related to the depth of the footing and overburden pressure. The third term is related to the width of the footing and the length of shear stress area. The bearing capacity factors, Nc, Nq, Nγ, are function of internal friction angle, φ.

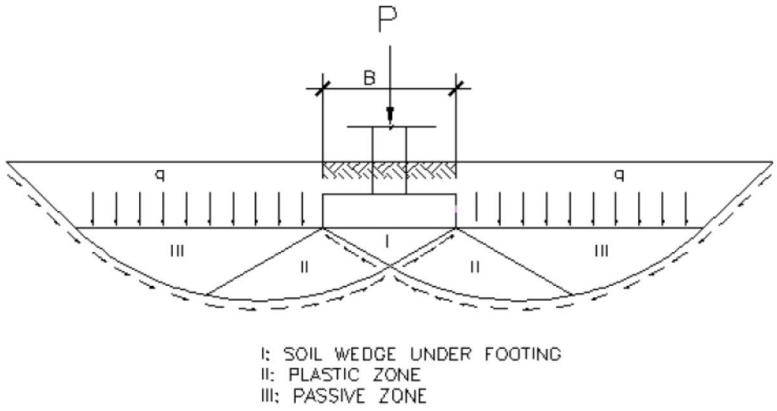


Figure 3.1: Shear stresses based on Terzaghi's soil bearing capacity theory

Terzaghi's Bearing capacity equations:

Strip footings:

$$Qu = c\,N_c + \gamma\,D\,N_q + 0.5\,\gamma\,B\,N_\gamma \quad \ldots\ldots (1)$$

Square footings:

$$Qu = 1.3\,c\,N_c + \gamma\,D\,N_q + 0.4\,\gamma\,B\,N_\gamma \quad \ldots\ldots (2)$$

Circular footings:

$$Qu = 1.3\,c\,N_c + \gamma\,D\,N_q + 0.3\,\gamma\,B\,N_\gamma \quad \ldots\ldots (3)$$

Where:

C: Cohesion of soil, γ: unit weight of soil, D: depth of footing, B: width of footing

N_c, N_q, N_r: Terzaghi's bearing capacity factors depend on soil friction angle, φ.

$N_c = \cot\varphi(N_q - 1)$

$Nq = e^{2(3\pi/4-\varphi/2)\tan\varphi} / [2\cos^2(45+\varphi/2)]$

$N_\gamma = (1/2)\tan\varphi(K_{p\gamma}/\cos^2\varphi - 1)$

$K_{p\gamma}$=passive pressure coefficient.

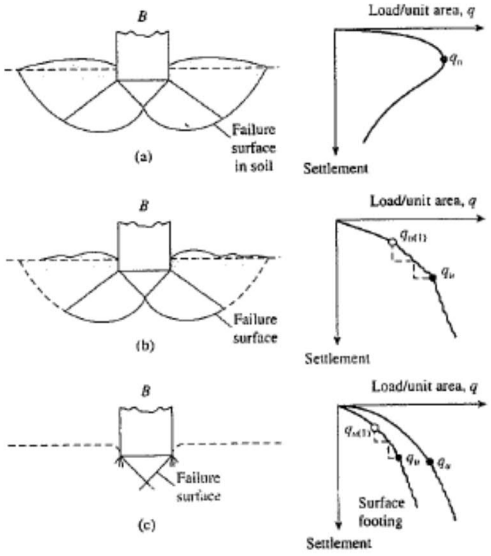


Figure 3.2 Nature of bearing capacity failure in soil:

(a) General Shear failure; (b) Local Shear Failure; (C) Punching Shear Failure (Redrawn after Vesic, 1973)

The above ultimate bearing capacity equations (1~3) are for continuous, square and circular foundations only; they do not address the case of rectangular foundations (0<B/L<1). Also the equations do not take into account the shearing resistance along the failure surface in soil above the bottom of the foundations. In addition, the load on the foundation may be inclined. To account for all these short comings, Meyerhof suggested that the following form of the general bearing capacity equation:

II. Meyerhof's General Bearing Capacity Equation

Vertical load:

$$Q_u = c\, N_c\, S_c\, D_c + \gamma\, D\, N_q\, S_q\, D_q + 0.5\, \gamma\, B\, N_\gamma\, S_\gamma\, D_\gamma$$

Inclined load:

$$Q_u = c\, N_c\, S_c\, D_c\, I_c + \gamma\, D\, N_q\, S_q\, D_q\, I_q + 0.5\, \gamma\, B\, N_\gamma\, S_\gamma\, D_\gamma\, I_\gamma$$

Where:

N_c, Nq, Nr	: Meyerhof's bearing capacity factors depend on soil friction angle, φ.
N_c	: cot φ (N_q – 1)
N_q	: $e^{\pi \tan\varphi} \tan^2 (45+\varphi/2)$]
N_γ	: (Nq-1) tan (1.4φ)
S_c, S_q, S_γ	: shape factors
D_c, D_q, D_γ	: depth factors
I_c, I_q, I_γ	: incline load factors
C	: Cohesion of soil
γ	: unit weight of soil
D	: depth of footing
B, L	: width and length of footing
Kp	: $\tan^2 (45+\varphi/2)$, passive pressure coefficient.
θ	: angle of axial load to vertical axis

Table 3.1: Meyerhof's bearing capacity factors

Friction angle	Shape factor	Depth factor	Incline load factors
Any ϕ	$Sc=1+0.2Kp(B/L)$	$Dc=1+0.2\sqrt{Kp}\ (B/L)$	$Ic=Iq=(1-\theta/90°)^2$
$\phi = 0$	$Sq=S\gamma=1$	$Dq=D\gamma=1$	$I\gamma= 0$
$\geq\phi 10°$	$Sq=S\gamma=1+0.1Kp(B/L)$	$Dq=D\gamma=1+0.1\sqrt{Kp}\ (D/B)$	$I\gamma=(1-\theta/\phi)^2$

Table 3.2: Meyerhof's bearing capacity factors

ϕ	Nc	Nq	Nr
0	5.1	1	0
5	6.5	1.6	0.1
10	8.3	2.5	0.4
15	11	3.9	1.2
20	14.9	6.4	2.9
25	20.7	10.7	6.8
30	30.1	18.4	15.1
35	46.4	33.5	34.4
40	75.3	64.1	79.4

Allowable soil bearing capacity shall be calculated taking into account the following safety factors.

$Q_a = Q_u / FOS$ (kN/m2)

Where;

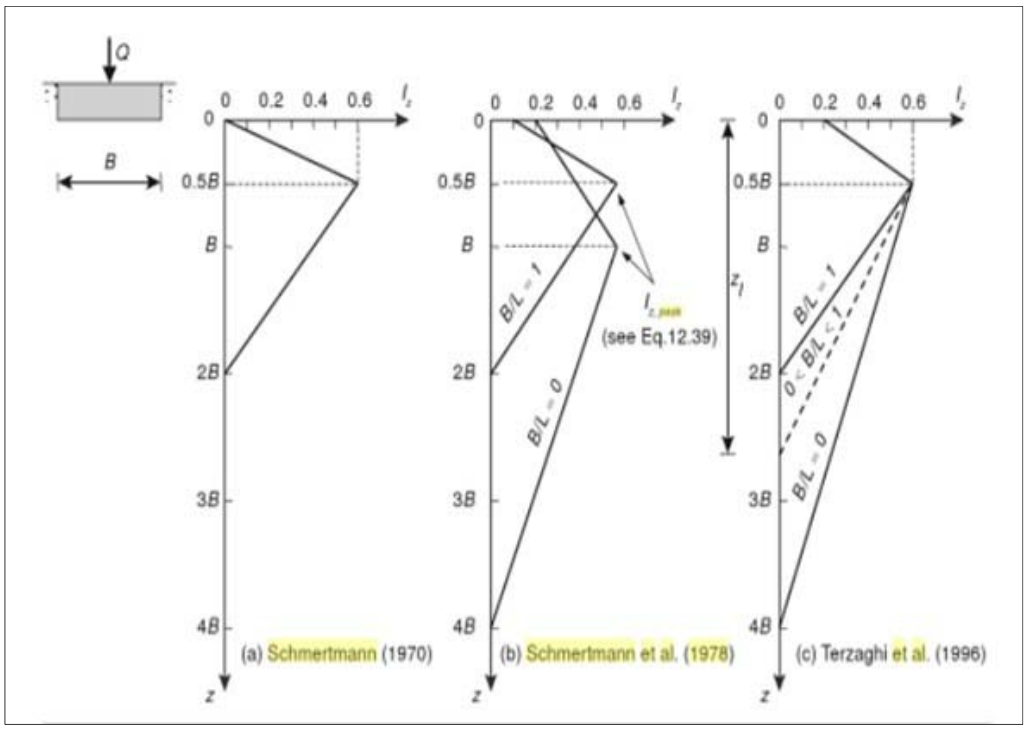
FOS = 3.0 : for permanent condition
FOS = 1.5 : for temporary condition

3.1.2. Settlement Calculation

For the analysis and computation of settlement in shallow foundations of cohesionless soils, the most popular method, Schmertmann's method (Schmertmann, 1970, 1978; and Schmertmann et al., 1978) is followed. This method is especially useful in complex soil profiles because it allows the engineer to divide the soil into layers and assign a different modulus to each layer. This method considers the relative importance of each layer through the use of strain influence factors.

Settlement calculation begins by performing appropriate in-situ tests to determine the modulus, Es, of the soil and its variation with depth. Settlement is then computed from the base of the footing to the depth of influence, or to an incompressible stratum, whichever is shallower.

The strain influence factor varies with depth, as shown in Fig. below



To compute the magnitude of this factor for each layer, it is first necessary to determine the peak value, I_{zp}:

$$I_{zp} = 0.5 + 0.1 \sqrt{(q' / \sigma'_{zp})}$$

Where,

I_{zp} : peak strain influence factor
q' : net bearing pressure (= $q - \sigma'_{vo}$)
σ'_{zp} : effective overburden pressure at foundation level (Depth = D_f + B/2 for square and circular footings, D + B for continuous footings)
z_f : depth below bottom of footing

And then compute the strain influence factor at the midpoint of each layer as follows:

$$\delta = C_1 \times C_2 \times C_3 \times q' \times \Sigma (I_{zi} \times H_i / E_{si})$$

Where,

δ : settlement of Footing
C_1 : depth factor = 1 - 0.5 ($\sigma'vo$ / q')
C_2 : creep factor = 1 + 0.2 log10 (time in years / 0.1)
C_3 : shape factor = 1.03 - 0.03 L/B $\geq$ 0.73
q' : net bearing pressure ($q - \sigma'_{vo}$)
H_i : thickness of soil layer I
E_{si} : modulus of soil layer i
I_{zi} : I_z at midpoint of soil layer i
σ'_{vo} : effective stress at a depth D_f below the ground surface

3.1.3. Allowable Settlement Criteria

Allowable settlement of foundation throughout the design life of the structure shall less than the values in the following table.

No.	Type	Allowable Settlement
1.	Spread Footing	25mm
2.	RC Mat Foundation	50mm

3.2. Analysis and Design of Piles

For the analysis and design of Piles, AASHTO LRFD Bridge Design Specifications (7th Edition) method is used. Its Design and analysis concept can be briefly presented in below.

3.2.1. Vertical Capacity of Pile

Ultimate capacity of the piles may be defined as the maximum load that a pile can carry before shear failure of the soil or excessive settlement takes place. The ultimate working load that can be applied to a given pile depends on the resistance that the pile can produce in terms of side friction and tip resistance.

The nominal axial compression resistance of a shaft is derived from the tip resistance and/or shaft side resistance, i.e., skin friction. Both the tip and shaft resistances develop in response to foundation displacement.

The factored resistance of drilled shafts, R_R, shall be taken as:

$$R_R = \varphi R_n = \varphi_{qp} R_p + \varphi_{qs} R_s$$

in which:

$$R_p = q_p A_p$$

$$R_S = q_S A_S$$

where:

R_p = nominal shaft tip resistance (kips)

R_s = nominal shaft side resistance (kips)

φ_{qp} = resistance factor for tip resistance specified in Table 10.5.5.2.4-1

φ_{qs} = resistance factor for shaft side resistance specified in Table 10.5.5.2.4-1

q_p = unit tip resistance (ksf)

q_s = unit side resistance (ksf)

A_p = area of shaft tip (ft^2)

A_s = area of shaft side surface (ft^2)

I) Estimation of Drilled Shaft Resistance in Cohesive Soils

Drilled shafts in cohesive soils should be designed by total and effective stress methods for undrained and drained loading conditions, respectively.

- **Side Resistance**

The nominal unit side resistance, qs, in ksf, for shafts in cohesive soil loaded under undrained loading conditions by the α-Method shall be taken as:

$$q_s = \alpha S_u$$

in which:

$$\alpha = 0.55 \quad \text{for} \quad \frac{S_u}{p_a} \le 1.5$$

$$\alpha = 0.55 - 0.1\left(S_u / p_a - 1.5\right)$$
$$\text{for} \quad 1.5 \le S_u / p_a \le 2.5$$

where:

S_u = undrained shear strength (ksf)

α = adhesion factor (dim)

p_a = atmospheric pressure (= 2.12 ksf)

- **Tip Resistance**

For axially loaded shafts in cohesive soil, the nominal unit tip resistance, qp, by the total stress method as provided in Brown et al. (2010) shall be taken as:

$$q_p = N_c S_u \leq 80.0 \text{ ksf}$$

in which:

$$N_c = 6\left[1 + 0.2\left(\frac{Z}{D}\right)\right] \leq 9$$

in which:

$$N_c = 6\left[1 + 0.2\left(\frac{Z}{D}\right)\right] \leq 9$$

where:

D = diameter of drilled shaft (ft)

Z = penetration of shaft (ft)

II) Estimation of Drilled Shaft Resistance in Cohesionless Soils

Shafts in cohesionless soils should be designed by effective stress methods for drained loading conditions or by empirical methods based on in-situ test results.

- **Side Resistance**
  The side resistance for shafts in cohesionless soils shall be determined using the β method, take as:

$$q_s = \beta\, \sigma'_v$$

in which:

$$\beta = \left(1 - \sin\varphi'_f\right)\left(\frac{\sigma'_p}{\sigma'_v}\right)^{\sin\varphi'_f} \tan\varphi'_f$$

where:

β = load transfer coefficient (dim)

σ'_f = friction angle of cohesionless soil layer (°)

σ'_p = effective vertical preconsolidation stress

σ'_v = vertical effective stress at soil layer mid-depth

The correlation for effective soil friction angle for use in the above equations shall be taken as:

$$\varphi'_f = 27.5 + 9.2\log\left[\left(N_1\right)_{60}\right] \qquad \text{(10.8.3.5.2b-3)}$$

where:

$(N_1)_{60}$ = SPT *N*-value corrected for effective overburden stress

The preconsolidation stress in Eq. 10.8.3.5.2b-2 should be approximated through correlation to SPT *N*-values. For sands:

$$\frac{\sigma'_p}{p_a} = 0.47\left(N_{60}\right)^m \qquad \text{(10.8.3.5.2b-4)}$$

where:

m = 0.6 for clean quartzitic sands
m = 0.8 for silty sand to sandy silts
p_a = atmospheric pressure (same units as σ'_p: 2.12 ksf or 14.7 psi)

For gravelly soils:

$$\frac{\sigma'_p}{p_a} = 0.15\left(N_{60}\right) \qquad \text{(10.8.3.5.2b-5)}$$

- **Tip Resistance**

The nominal tip resistance, qp, in ksf, for drilled shafts in cohesionless soils by the method described in Brown et al. (2010) shall be taken as:

If $N_{60} \le 50$, then $q_p = 1.2N_{60}$

where:

N_{60} = average *SPT* blow count (corrected only for hammer efficiency) in the design zone under consideration (blows/ft)

The value of q_p in Eq. 10.8.3.5.2c-1 should be limited to 60 ksf, unless greater values can be justified though load test data.

III) Uplift Resistance

Uplift resistance shall be evaluated when upward loads act on the drilled shafts. Drilled shafts subjected to uplift forces shall be investigated for resistance to pullout, for their structural strength, and for the strength of their connection to supported components.
The factored nominal uplift resistance of a belled drilled shaft in a cohesive soil, RR, in kips, should be determined as:

$$R_R = \varphi R_n = \varphi_{up} R_{sbell}$$

in which:

$$R_{s\ bell} = q_{s\ bell} A_u$$

where:

q_{sbell} = $N_u S_u$ (ksf)

A_u = $\pi(D_p^2 - D^2)/4$ (ft^2)

N_u = uplift adhesion factor (dim)

D_p = diameter of the bell (ft)

D_b = depth of embedment in the founding layer (ft)

D = shaft diameter (ft)

S_u = undrained shear strength averaged over a distance of 2.0 bell diameters ($2D_p$) above the base (ksf)

φ_{up} = resistance factor specified in Table 10.5.5.2.4-1

3.2.2. Lateral Capacity of Pile

Lateral analysis has been performed using AllPile software. AllPile directly uses COM624S calculation methods for lateral analysis. For details on COM624, please refer to the FHWA publications, FHWA-SA-91-048, COM624P–Laterally Loaded Pile Program for the Microcomputer, Version 2.0, by Wang and Reese (1993).
A pile loaded by lateral thrust and/or moment at its top, resists the load by deflecting to mobilize the reaction of the surrounding soil. The magnitude and distribution of the resisting pressures are a function of the relative stiffness of pile and soil.

Design criteria is based on maximum combined stress in the piling, allowable deflection at the top or permissible bearing on the surrounding soil.

Below is brief introduction to the program. COM624S uses the four nonlinear differential equations to perform the lateral analysis.

Lateral Deflection Calculation

Here is brief introduction to the program. COM624S uses the four nonlinear differential equations to perform the lateral analysis. They are:

$$EI\frac{d^4Y}{dZ^4}+Q\frac{d^2Y}{dZ^2}-R-P_q=0 \qquad (1)$$

Where Q = axial compression load on the pile

Y = lateral deflection of pile at depth of Z

Z = depth from top of pile

R = soil reaction per unit length

E = modules of elasticity of pile

I = moment of inertia of the pile

P_q = distributed load along the length of pile

$$EI(\frac{d^3Y}{dZ})+Q(\frac{dY}{dZ})=P \qquad (2)$$

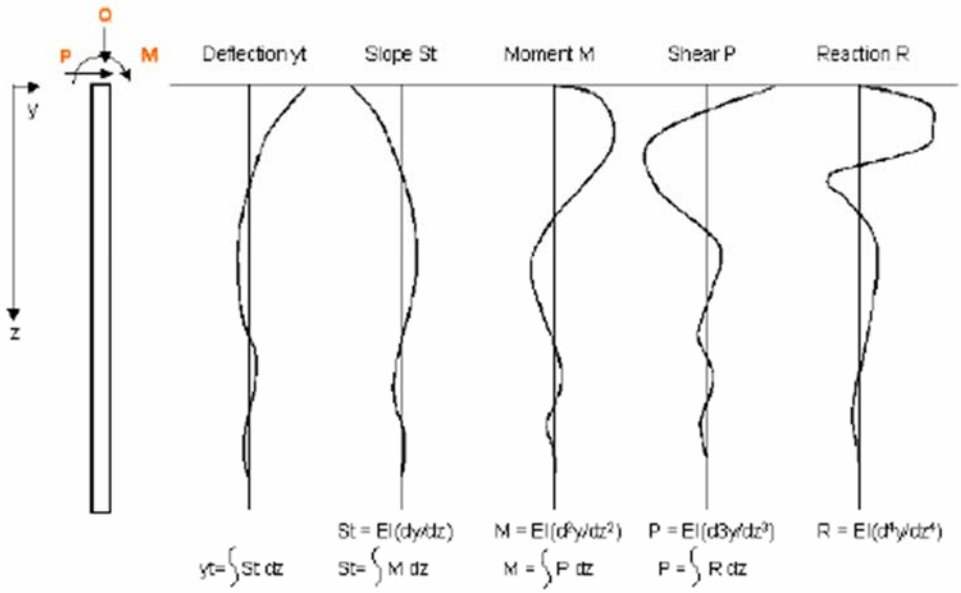
Where P = shear in the pile

Where M = bending moment of the pile

$$\frac{dY}{dZ}=S_t \qquad (4)$$

$$EI(\frac{d^2Y}{dZ^2})=M \qquad (3)$$

Where S_t = slope of the elastic curve defined by the axis of pile



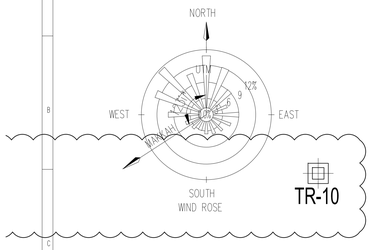
4. SUMMARY OF SHALLOW FOUNDATION ANALYSIS RESULT

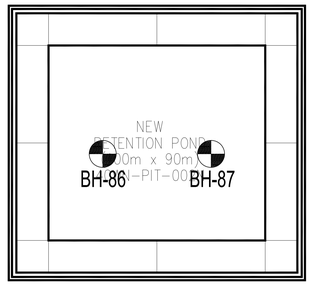
4.1 Bore Hole Location For Shallow Foundation Map

SHALLOW FOUNDATION CONDITION

NOTE

1. ALL COORDINATES ARE IN METERS.
2. PLEASE REFER TO GEOTECHNICAL INVESTIGATION PLAN_P02 FOR TR-07, TR-08, TR-09, TR-10





BATTERY LIMIT (UTM N. 3,046,698.000)

BATTERY LIMIT E. 322,708.000

BATTERY LIMIT E. 323,488.030

BATTERY LIMIT N. 3,046,200.000

FUTURE TANKAGE AREA

MAIN GATE

BH-93 BH-151 BH-94 BH-95 BH-96 BH-97 BH-98 BH-99 BH-100 BH-101 BH-102 BH-103 BH-104 BH-105 BH-106 BH-107 BH-108 BH-109 BH-110 BH-111 BH-112 BH-113 BH-114 BH-115 BH-116 BH-117 BH-118 BH-119 BH-120 BH-121 BH-122 BH-123 BH-124 BH-125 BH-126 BH-127 BH-128 BH-129 BH-130 BH-131 BH-132 BH-133 BH-134 BH-135 BH-136 BH-137 BH-138 BH-139 BH-140 BH-141 BH-142 BH-143 BH-144 BH-145 BH-146

BH-07 BH-148

BH-50

OF ROAD (UTM N. 3,046,598.000)

OF ROAD (UTM N. 3,046,484.000)

OF ROAD (UTM N. 3,046,443.000)

OF ROAD (UTM N. 3,046,298.000)

OF ROAD (UTM E. 322,731.000)

OF ROAD (UTM E. 323,027.500)

OF ROAD (UTM E. 323,154.000)

OF ROAD (UTM E. 323,294.000)

OF ROAD N. 3,046,338.000

OF ROAD N. 3,046,308.000

OF ROAD N. 3,046,211.000

CPT-01 CPT-02 CPT-03 CPT-04 CPT-05 CPT-06 CPT-07 CPT-08

ER-01 ER-02 ER-03 ER-04 ER-05 ER-06 ER-07 ER-08 ER-09 ER-10 ER-11 ER-12 ER-13 ER-14 ER-15 ER-16 ER-17 ER-18 ER-19

TR-01 TR-02 TR-03 TR-04 TR-05 TR-06

DHT-01 DHT-02 DHT-03 DHT-04 DHT-05 DHT-06 DHT-07 DHT-08

CHT-01 CHT-02 CHT-03 CHT-04

NEW IRRIGATION POND 45m x 45m BH-84 BH-85

BH-20

LEGEND

- : BOREHOLES LOCATION (BH) - 148 NOS
- : CROSSHOLE TEST (CHT) - 4 NOS
- : DOWNHOLES TEST (DHT) - 8 NOS
- : CONE PENETRATION TEST (CPT) - 8 NOS
- : ELECTRICAL RESISTIVITY TEST (ER) - 19 NOS
- : THERMAL CONDUCTIVITY TEST (TR) - 10 NOS

METERS 0 20 40 60 80 100 120

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Result	Code	Description
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☐	C	Proceed with exceptions to next submission and Revision
☐	R	Do Not Proceed – Revise as noted and Resubmit

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PROJECT MANAGER SIGN-OFF:

NAME: DATE:

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
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DAELIM JACOBS MA'ADEN معادن Saudi Arabian Mining Company

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CHECKED	–	–	
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APPROVED	–	–	
EMPLOYER			
SCALE	1:1000	A0	

MA'ADEN AMMONIA PROJECT

ATTACHMENT 5

GEOTECHNICAL INVESTIGATION PLAN (1/2)

DRAWING NUMBER –

REV P06

4.2 Shallow Foundation Analysis Result Table

Table 1.1 Summary of Soil Contact Pressure & Soil Bearing Capacity (Tank Area)

FDN Condition			Contract Pressure of 25mm Settlement Calculated by Schmertmann Method(KPa)
	Width (m)	Depth (m)	BH-07
	3*3	2	94

FDN Condition			Contract Pressure by Meyerhof's Method(KPa)
	Width (m)	Depth (m)	BH-07
	3*3	2	622

Table 1.2 Summary of Soil Contact Pressure & Soil Bearing Capacity (Utility Area)

FDN Condition			Contract Pressure of 25mm Settlement Calculated by Schmertmann Method(KPa)
	Width (m)	Depth (m)	BH-50
	3*3	2	110

FDN Condition			Contract Pressure by Meyerhof's Method(KPa)
	Width (m)	Depth (m)	BH-50
	3*3	2	948

5. SUMMARY OF PILE FOUNDATION ANALYSIS RESULT

5.1 Bore Hole Location For Pile Foundation Map

DRAWING NUMBER –
REV P06

NOTE

1. ALL COORDINATES ARE IN METERS.
2. PLEASE REFER TO GEOTECHNICAL INVESTIGATION PLAN_P02 FOR TR-07, TR-08, TR-09, TR-10

NORTH
WEST
EAST
SOUTH
WIND ROSE
TR-10

NEW RETENTION POND 90m x 90m
BH-86 BH-87

BATTERY LIMIT (UTM N. 3,046,698.000)

BH-94 BH-96 BH-98 BH-100 BH-102 BH-104 BH-106 BH-107 BH-108 BH-109
BH-97 BH-99 BH-101 BH-103 BH-105
FUTURE TANKAGE AREA
MAIN GATE
BH-95
BH-93 BH-151
℄ OF ROAD (UTM N. 3,046,598.000)
BH-01 BH-07 BH-138
BH-126 CPT-01 BH-04 BH-147 BH-05 T1-15A BH-10 BH-149 BH-11
BH-06 BH-12
BH-02 BH-03 ER-01 BH-150 BH-139
BH-13 BH-14 BH-15 BH-16 DHT-01 BH-17 BH-18 BH-19
℄ OF ROAD (UTM N. 3,046,484.000)
ER-02 BH-127
BH-128 BH-129 BH-130 CPT-02
BH-110 BH-111 BH-112 BH-113
BH-114 BH-115 BH-116 BH-117
BH-118 BH-119 BH-120 BH-121
BH-122 BH-123 BH-124 BH-125
℄ OF ROAD (UTM N. 3,046,443.000)
BH-21 BH-22 BH-23 BH-24
CPT-04 ER-04 BH-25 TR-03 CPT-06 BH-64 BH-65 CPT-07 BH-134
BH-131 BH-26 BH-27 BH-28 BH-29 BH-30 BH-31 CPT-03 BH-49 BH-52 BH-53 BH-59 ER-13 BH-60 BH-66 BH-69 BH-70 BH-153 CPT-08
U1-16
CHT-01 DHT-02 TR-01 BH-33 BH-34 BH-35 ER-05 DHT-03 BH-36 DHT-04 BH-37 ER-03 DHT-06 BH-62 BH-61 TR-04 ER-14 ER-16 BH-152 BH-135
BH-132 CHT-02 CHT-03 CHT-04 TR-02 BH-50 BH-54 BH-63
ER-06 BH-38 BH-39 BH-40 BH-41 BH-42 BH-51 BH-55 BH-71 BH-72 BH-73
BH-32
BH-133 ER-07 ER-11
BH-67 ER-15 BH-68
ER-08 DHT-05 BH-56 BH-57 BH-58 DHT-07
BH-43 ER-09 ER-10
BH-44 BH-45 BH-46 BH-47 BH-48 ER-12
℄ OF ROAD (UTM N. 3,046,298.000)
℄ OF ROAD N. 3,046,338.000
℄ OF ROAD N. 3,046,308.000
BH-77 BH-78 BH-140 BH-141 BH-142
BH-136 BH-146 BH-76 TR-05 BH-81 TR-06
BH-74 BH-75 ER-18 BH-80 BH-82
BH-137 CPT-05 BH-79 ER-19 BH-20 BH-143 BH-144 BH-145
ER-17 DHT-08 BH-83
NEW IRRIGATION POND 45m x 45m
BH-84 BH-85

BATTERY LIMIT E. 322,708.000
BATTERY LIMIT E. 323,488.030
BATTERY LIMIT N. 3,046,200.000

LEGEND

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- : CROSSHOLE TEST (CHT) - 4 NOS
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- : CONE PENETRATION TEST (CPT) - 8 NOS
- : ELECTRICAL RESISTIVITY TEST (ER) - 19 NOS
- : THERMAL CONDUCTIVITY TEST (TR) - 10 NOS

METERS
0 20 40 60 80 100 120

DOCUMENT APPRAISAL STAMP

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REVIEWER'S NAME	SIGN	DATE

REVIEW GRADE

Result	Code	Description
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NAME: DATE:

$FILE$ $TIME$ $DATE$

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ENGINEER	–	–
APPROVED	–	–
EMPLOYER		
SCALE	1:1000	A0

MA'ADEN AMMONIA PROJECT
ATTACHMENT 5
GEOTECHNICAL INVESTIGATION PLAN (1/2)
DRAWING NUMBER –
REV P06

5.2 Pile Capacity Analysis Result Table

Table 1.1 Summary of Allowable Pile Capacity of TANK Area

TANK (FDN Bottom -1m)

AREA	BH #	Pile Dia (mm)	Pile Tip EL. (m)	Pile Cutoff Level (P.C.O)	Soil Depth over P.C.O (m)	Allowable Capacity (kN)						Applied Capacity (kN)		
						Compression (Allowable)	Compression (Material)	Tension (Allowable)	Tension (Material	Lateral (Allowalbe)	Lateral (Material	Compression	Tension	Lateral
T1-15A	BH-07	800	-13	+1.9m	0	1900	2000	1060	650	130	130	1900	650	130

Table 1.2 Summary of Allowable Pile Capacity of UTILITY Area

UTILITY (FDN Bottom -1m)

AREA	BH #	Pile Dia (mm)	Pile Tip EL. (m)	Pile Cutoff Level (P.C.O)	Soil Depth over P.C.O (m)	Allowable Capacity (kN)						Applied Capacity (kN)		
						Compression (Allowable)	Compression (Material)	Tension (Allowable)	Tension (Material	Lateral (Allowalbe)	Lateral (Material	Compression	Tension	Lateral
U1-15	BH-50	600	-13	-1m	1m	1300	1500	700	300	110	110	1300	300	110

*Group Effect Coefficient

- AASHTO LRFD Bridge Design Specification (2014)
- Table 10.8.3.6.3-1 Group Reduction Factors for Bearing Resistance of Shafts in Sand

Table 10.8.3.6.3-1—Group Reduction Factors for Bearing Resistance of Shafts in Sand

Shaft Group Configuration	Shaft Center-to-Center Spacing	Special Conditions	Reduction Factor for Group Effects, η
Single Row	2*D*		0.90
	3*D* or more		1.0
Multiple Row	2.5*D*		0.67
	3*D*		0.80
	4*D* or more		1.0
Single and Multiple Rows	2*D* or more	Shaft group cap in intimate contact with ground consisting of medium dense or denser soil, and no scour below the shaft cap is anticipated	1.0
Single and Multiple Rows	2*D* or more	Pressure grouting is used along the shaft sides to restore lateral stress losses caused by shaft installation, and the shaft tip is pressure grouted	1.0

Appendix

I) Sample Shallow Foundation Calculation sheet

SHALLOW FOUNDATION ANALYSIS RESULT

BH No.	Proposed Structure	Foundation Size (B x L) (m)		Footing Depth (m)	Meyerhof's Bearing capacity (kPa)	Schemertmann's Settlement Analysis		Remark
						Applied Pressure (kPa)	Induced Settlement (mm)	
BH-7	Tank Area	3	3	2	622	94	24.83	

BH	Layer Description	Site Parameters				Soil Parameters							
		Ground Elevation (E.L. +/-) (m)		Ground Water Level (m)		Depth (m)		Y_t kN/m^3	C (kPa)	Phi (^{O})	N Value (SPT)	E' (MPa)	Applied E kN/m^2
		N.G.L.	F.G.L.	N.G.L. (-)	F.G.L. (-)	Below F.G.L.	E.L. (+/-)					Calculated as per agreed MOM	
BH-7	Fill Area	#########	########	#########	F.G.L - 5.00	0.00	7.00	17.5	0	31.0	10/30	10.0	10,000
	Sabkha					1.00	6.00	17.0	0	27.0	4/30	3.0	3,000
	Loose Sand					1.75	5.25	17.5	0	31.0	7/30	7.0	7,000
	Loose Sand					2.50	4.50	17.5	0	31.0	8/30	8.0	8,000
	Loose Sand					3.25	3.75	17.5	0	31.0	10/30	10.0	10,000
	Sabkha					4.00	3.00	17.0	0	27.0	3/30	2.7	2,700
	Sabkha					4.75	2.25	17.0	0	27.0	4/30	3.0	3,000
	Sabkha					5.50	1.50	17.0	0	27.0	2/30	2.4	2,400
	Dense Sand					7.00	0.00	18.5	0	36.0	34/30	34.0	34,000
	Dense Sand					8.50	-1.50	18.5	0	36.0	47/30	47.0	47,000
	Dense Sand					10.00	-3.00	18.5	0	36.0	50/30	50.0	50,000
	Medium Dense Sand					11.50	-4.50	18.0	0	34.0	30/30	30.0	30,000
	Dense Sand					13.00	-6.00	18.5	0	36.0	34/30	34.0	34,000
	Medium Dense Sand					14.50	-7.50	18.0	0	34.0	26/30	26.0	26,000
	Dense Sand					16.00	-9.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					17.50	-10.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					19.00	-12.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					20.50	-13.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					22.00	-15.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					23.50	-16.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					25.00	-18.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					26.50	-19.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					28.00	-21.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					29.50	-22.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					31.00	-24.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					32.50	-25.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					34.00	-27.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					35.50	-28.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					37.00	-30.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					38.50	-31.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					40.00	-33.00	18.5	0	36.0	50/30	50.0	50,000

Bearing Capacity Calculation For BH-7

(A) Input Parameters

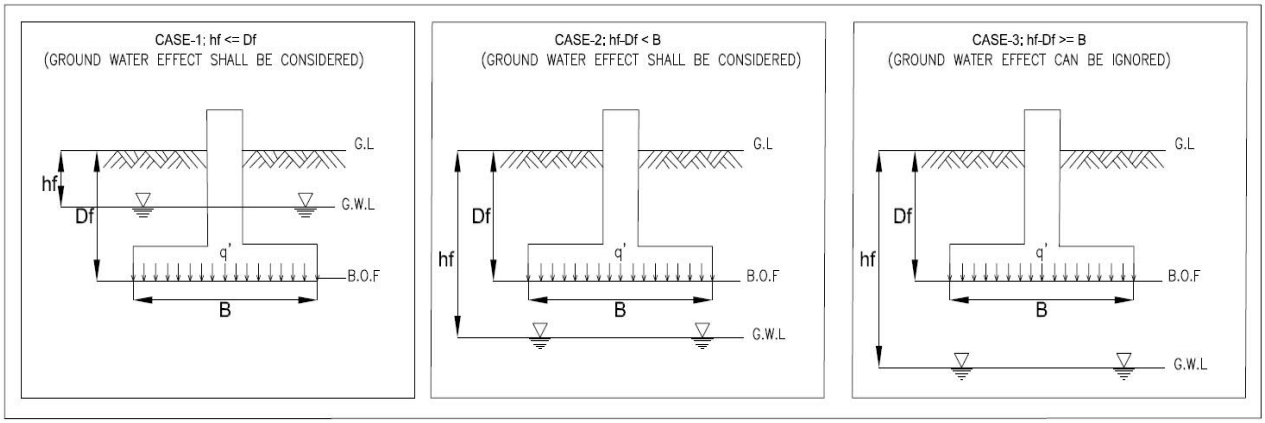
■Soil Parameter

Φ	c	Unit Wt(Above Base of Fdn.) γ_1	Unit Wt(Below Base of Fdn.) γ_2	Unit Wt of Water γ_w	G.W.L (h_f)
degree	kPa	kN/m^3	kN/m^3	kN/m^3	GL(-), m
31	0.00	17.50	17.50	10.00	5.00

■Foundation Dimension

Width, B	Eff. Width B'	Length, L	Eff. Length L'	Depth of Footing. D_f	Soil Slope	Fdn. Slope	Eff. Area, A_f
m	m	m	m	m	β, degree	η,degree	m^2
3	3.0	3	3	2	0	0	9.00

■Ground water cases: the Fig below shows different cases of ground water location.



- General Formula:

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma_2 \cdot B \cdot Nr \cdot Sr \cdot dr$

- The unit weights in the second and third term of the above equation shall be modified depending on the location of water table as explained in below.

- Case-1: hf<=Df

$q_u = c \cdot N_c \cdot S_c \cdot d_c + ((hf \cdot \gamma_1)+(Df-hf) \cdot (\gamma')) \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot (\gamma'_2) \cdot B \cdot Nr \cdot Sr \cdot dr \quad ; \quad \gamma'=\gamma-\gamma_w$

- Case-2: hf-Df<B

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot [(\gamma') + ((hf-Df)/B) \cdot (\gamma-\gamma'))] \cdot B \cdot Nr \cdot Sr \cdot dr$

- Case-3: hf-Df>=B

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma 2 \cdot B \cdot Nr \cdot Sr \cdot dr$

∴ Ground water case is: CASE-3

(B) Meyerhof's Method of Bearing Capacity Calculation

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma_2 \cdot B \cdot Nr \cdot Sr \cdot dr$

① Nc, Nq, Nr : Bearing capacity factor

Nc = 33.30 ; Nr = 19.98 ; Nq = 21.38

② Shape factor

Sc = 1.62 ; Sr = 1.31 ; Sq = 1.31 Kp = 3.12

Kp = TAN((45+Φ/2)*PI()/180)^2
Sc = 1+0.2*Kp*(B/L)
Sr = 1+0.1*Kp*(B/L)
Sq = Sr

③ **Depth factor**

dc = 1.24 ; dr = 1.12 ; dq = 1.12

dc = 1+0.2*(Kp^0.5)*(Df/L)
dr = 1+0.1*(Kp^0.5)*(Df/L)
dq = dr

■ Meyrhof's Bearing Capacity Factors

Angle	Nc	Nr	Nq
0	5.14	0	1
5	6.5	0.1	1.6
10	8.3	0.4	2.5
15	11	1.1	3.9
20	14.8	2.9	6.4
25	20.7	6.8	10.7
30	30.1	15.7	18.4
35	46.1	37.1	33.3
40	75.3	93.7	64.2
45	133.9	262.7	134.9
50	266.9	873.7	319

■ Effective Unit Weights

Item	Cases	Applied Formula	Value	Use
q'=	Case-1	$hf*\gamma_1 + (Df-hf)*\gamma'_1$	-	35
	Case-2 & 3	$Df*\gamma_1$	35	
γ'_2=	Case-1	$\gamma_2-\gamma_w$	-	17.5
	Case-2	$((\gamma_2-\gamma_w) +((hf-Df)/B) \cdot (\gamma_2-(\gamma_2-\gamma_w)))$	-	
	Case-3	γ_2	17.5	

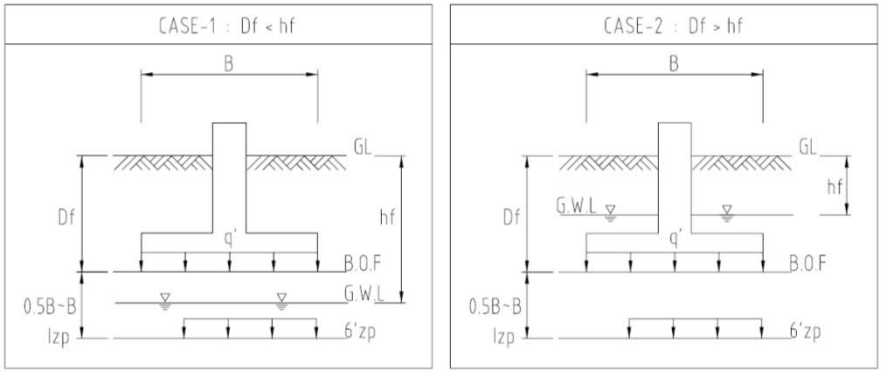
$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma_2 \cdot B \cdot Nr \cdot Sr \cdot dr$
$= 0.00 + 1097.79 + 769.43$
$= 1867.22$ kPa
$q_a = 1867.22 / 3$
$q_a =$ **622.41** (kPa) ; is the the Allowabel Bearing Capacity (kPa)

2. Settlement Calculation For BH-7

■ Footing dimension

Width (B)	Length (L)	Depth (D_f)	h_f	L/B	Footing Shape	Unit Wt of Soil
3.00 m	3.00 m	2.00 m	5.00 m	1	SQU	18 kN/m3

■ Footing shape
L/B = 1, : Square (SQU) and Circular Footings (CIR)
L/B ≥ 10, : Continuous Footings (CON)
1< L/B < 10, : Rectangular Footings (REC)



■ Depth of influence = 8.0 m
where, Square Shape = Df+2B = 8.0 m
Continuous Shape = Df+4B = 14.0 m
* For Rec Shape it shall be determind By linear Interpolation of Square and Contineous Types -

■ Effective stress at a depth D_f below the ground surface (σ'vo)

- σ'vo = 35.00 kN/m^2

where, For Case-1 : $\sigma'_{vo} = (\gamma_s \times D_f)$
For Case-2 : $\sigma'_{vo} = (\gamma_s \times h_f) + (\gamma_s - \gamma_w) \times (D_f - h_f)$

■ Initial vertical effective stress at a depth of I_{zp} (σ'_{zp})

- σ'_{zp} = 61.25 kN/m^2

- $\sigma'_{zp(squ)}$ = 61.25 kN/m^2 : (For Square)

where, $\sigma'_{zp(squ)} = (\gamma_s \times h_f) + (\gamma_s - \gamma_w) \times (D_f + B/2 - h_f)$: For $h_f < D_f + B/2$
$\sigma'_{zp(squ)} = \gamma_s \times (D_f + B/2)$: For $h_f > D_f + B/2$

- $\sigma'_{zp(rec)}$ = - kN/m^2 : (Rec. Shape Footing)
* For Rec Shape it shall be determind By linear Interpolation of Square and Contineous Types

- $\sigma'_{zp(con)}$ = 87.50 kN/m^2 : (For Continuous Shape Footing)

where, $\sigma'_{zp(con)} = (\gamma_s \times h_f) + (\gamma_s - \gamma_w) \times (D_f + B - h_f)$: For $h_f < D_f + B$
$\sigma'_{zp(con)} = \gamma_s \times (D_f + B)$: For $h_f > D_f + B$

■ Bearing pressure at bottom of footing (q)

- q = 94.00 kN/m^2

■ Peak strain influence factor (I_{zp}) = 0.598

- $I_{zp(squ)} = 0.5 + 0.1 \sqrt{(q' / \sigma'_{zp(squ)})}$ = 0.60
- $I_{zp(con)} = 0.5 + 0.1 \sqrt{(q' / \sigma'_{zp(con)})}$ = 0.58
* For Rec Shape it shall be determind By linear Interpolation of Square and Contineous Types -

■ Depth factor (C_1)

- $C_1 = 1 - 0.5 (\sigma'_{vo} / q')$
  $= 0.703$

■ Secondary creep factor (C_2)

- $C_2 = 1 + 0.2 \log_{10} (t / 0.1)$
  $= 1.400$

■ Time since application of load (yr)

- $t =$ 10.0 year ($t \geq 0.1$ yr)

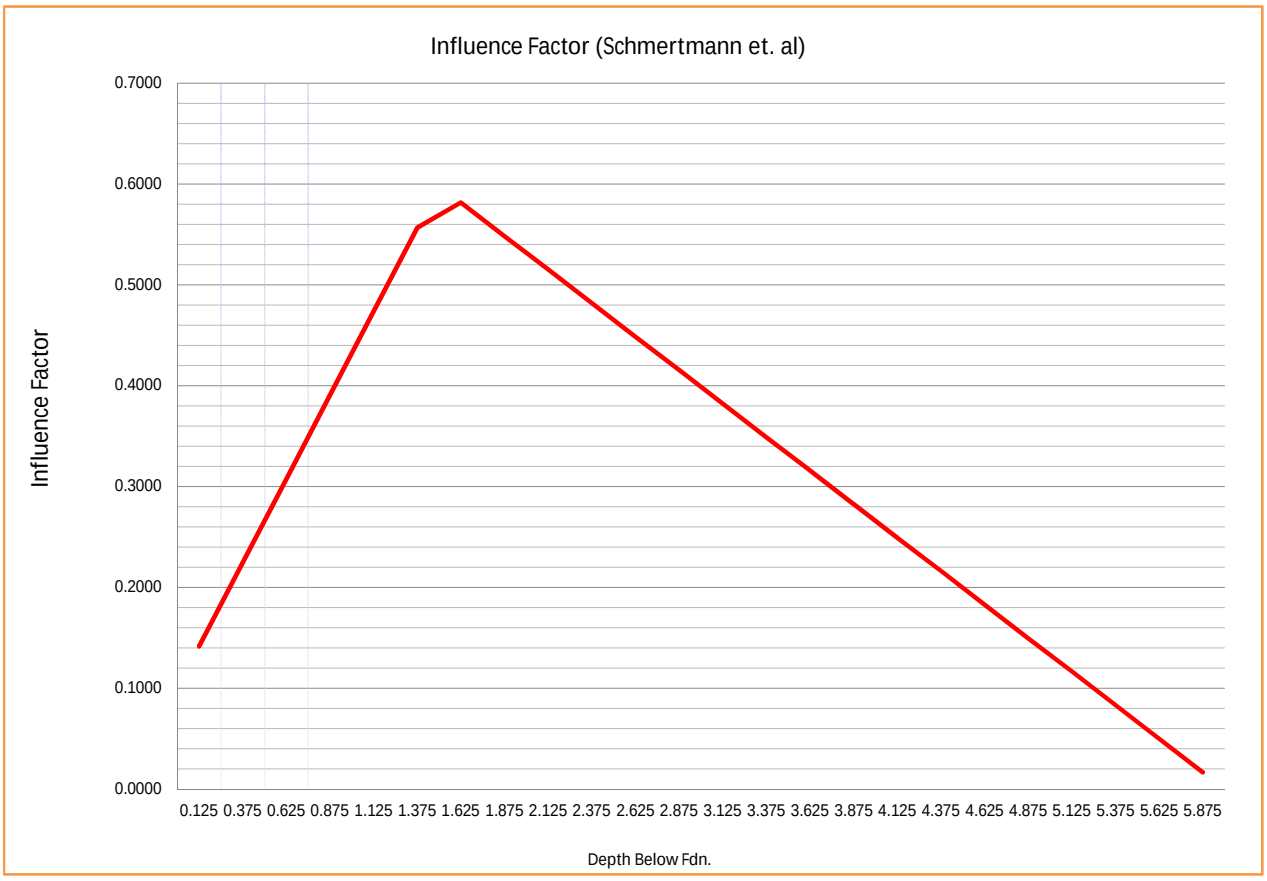
■ Shape factor (C_3)

- $C_3 = 1.03 - 0.03 (L / B) \geq 0.73$
  $= 1.000$

Depth interval = 0.25 m

Top (m)	Bottom (m)	E_s (kN/m^2)	z_f (m)	$I_{z(squ)}$	$I_{z(rec)}$	$I_{z(con)}$	I_{zi}	H_i (m)	$I_{zi} H_i / E_s$	**δ** (mm)	
0.0	2.00										
2.0	2.25	7,000	0.125	0.1415	NA	NA	0.1415	0.25	5.05E-06	0.2936	
2.3	2.50	8,000	0.375	0.2245	NA	NA	0.2245	0.25	7.02E-06	0.4077	
2.5	2.75	8,000	0.625	0.3076	NA	NA	0.3076	0.25	9.61E-06	0.5584	
2.8	3.00	8,000	0.875	0.3906	NA	NA	0.3906	0.25	1.22E-05	0.7092	
3.0	3.25	10,000	1.125	0.4736	NA	NA	0.4736	0.25	1.18E-05	0.6879	
3.3	3.50	10,000	1.375	0.5566	NA	NA	0.5566	0.25	1.39E-05	0.8085	
3.5	3.75	10,000	1.625	0.5815	NA	NA	0.5815	0.25	1.45E-05	0.8447	
3.8	4.00	2,700	1.875	0.5483	NA	NA	0.5483	0.25	5.08E-05	2.9497	
4.0	4.25	2,700	2.125	0.5151	NA	NA	0.5151	0.25	4.77E-05	2.7709	
4.3	4.50	2,700	2.375	0.4818	NA	NA	0.4818	0.25	4.46E-05	2.5921	
4.5	4.75	3,000	2.625	0.4486	NA	NA	0.4486	0.25	3.74E-05	2.1720	
4.8	5.00	3,000	2.875	0.4154	NA	NA	0.4154	0.25	3.46E-05	2.0111	
5.0	5.25	3,000	3.125	0.3821	NA	NA	0.3821	0.25	3.18E-05	1.8502	
5.3	5.50	2,400	3.375	0.3489	NA	NA	0.3489	0.25	3.63E-05	2.1117	
5.5	5.75	2,400	3.625	0.3157	NA	NA	0.3157	0.25	3.29E-05	1.9106	
5.8	6.00	2,400	3.875	0.2825	NA	NA	0.2825	0.25	2.94E-05	1.7095	
6.0	6.25	34,000	4.125	0.2492	NA	NA	0.2492	0.25	1.83E-06	0.1065	
6.3	6.50	34,000	4.375	0.2160	NA	NA	0.2160	0.25	1.59E-06	0.0923	
6.5	6.75	34,000	4.625	0.1828	NA	NA	0.1828	0.25	1.34E-06	0.0781	
6.8	7.00	34,000	4.875	0.1495	NA	NA	0.1495	0.25	1.10E-06	0.0639	
7.0	7.25	34,000	5.125	0.1163	NA	NA	0.1163	0.25	8.55E-07	0.0497	
7.3	7.50	34,000	5.375	0.0831	NA	NA	0.0831	0.25	6.11E-07	0.0355	
7.5	7.75	47,000	5.625	0.0498	NA	NA	0.0498	0.25	2.65E-07	0.0154	
-End of Influence- 7.8	8.00	47,000	5.875	0.0166	NA	NA	0.0166	0.25	8.84E-08	0.0051	

Σtotal δ* = 25 mm


Influence Factor (Schmertmann et. al)
Influence Factor
0.7000
0.6000
0.5000
0.4000
0.3000
0.2000
0.1000
0.0000
0.125 0.375 0.625 0.875 1.125 1.375 1.625 1.875 2.125 2.375 2.625 2.875 3.125 3.375 3.625 3.875 4.125 4.375 4.625 4.875 5.125 5.375 5.625 5.875
Depth Below Fdn.

SHALLOW FOUNDATION ANALYSIS RESULT

BH No.	Proposed Structure	Foundation Size (B x L) (m)		Footing Depth (m)	Meyerhof's Bearing capacity (kPa)	Schemertmann's Settlement Analysis		Remark
						Applied Pressure (kPa)	Induced Settlement (mm)	
BH-50	Utility Area	3	3	2	948	110	24.30	

BH	Layer Description	Site Parameters				Soil Parameters							
		Ground Elevation (E.L. +/-) (m)		Ground Water Level (m)		Depth (m)		Y_t kN/m^3	C (kPa)	Phi (^{O})	N Value (SPT)	E' (MPa)	Applied E kN/m^2
		N.G.L.	F.G.L.	N.G.L. (-)	F.G.L. (-)	Below F.G.L.	E.L. (+/-)					Calculated as per agreed MOM	
BH-50	Fill Area	#########	########	#########	F.G.L - 5.00	0.00	7.00	17.5	0	31.0	10/30	10.0	10,000
	Loose Sand					1.00	6.00	17.5	0	31.0	6/30	6.0	6,000
	Sabkha					1.75	5.25	17.0	0	27.0	5/30	3.3	3,000
	Sabkha					2.50	4.50	17.0	0	27.0	5/30	3.3	3,000
	Medium Dense Sand					3.25	3.75	18.0	0	34.0	13/30	13.0	13,000
	Medium Dense Sand					4.00	3.00	18.0	0	34.0	22/30	22.0	22,000
	Medium Dense Sand					4.75	2.25	18.0	0	34.0	13/30	13.0	13,000
	Sabkha					5.50	1.50	17.0	0	27.0	1/30	2.1	2,100
	Medium Dense Sand					7.00	0.00	18.0	0	34.0	19/30	19.0	19,000
	Dense Sand					8.50	-1.50	18.5	0	36.0	32/30	32.0	32,000
	Medium Dense Sand					10.00	-3.00	18.0	0	34.0	25/30	25.0	25,000
	Dense Sand					11.50	-4.50	18.5	0	36.0	34/30	34.0	34,000
	Dense Sand					13.00	-6.00	18.5	0	36.0	39/30	39.0	39,000
	Dense Sand					14.50	-7.50	18.5	0	36.0	36/30	36.0	36,000
	Dense Sand					16.00	-9.00	18.5	0	36.0	48/30	48.0	48,000
	Dense Sand					17.50	-10.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					19.00	-12.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					20.50	-13.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					22.00	-15.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					23.50	-16.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					25.00	-18.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					26.50	-19.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					28.00	-21.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					29.50	-22.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					31.00	-24.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					32.50	-25.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					34.00	-27.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					35.50	-28.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					37.00	-30.00	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					38.50	-31.50	18.5	0	36.0	50/30	50.0	50,000
	Dense Sand					40.00	-33.00	18.5	0	36.0	50/30	50.0	50,000

Bearing Capacity Calculation For BH-50

(A) Input Parameters

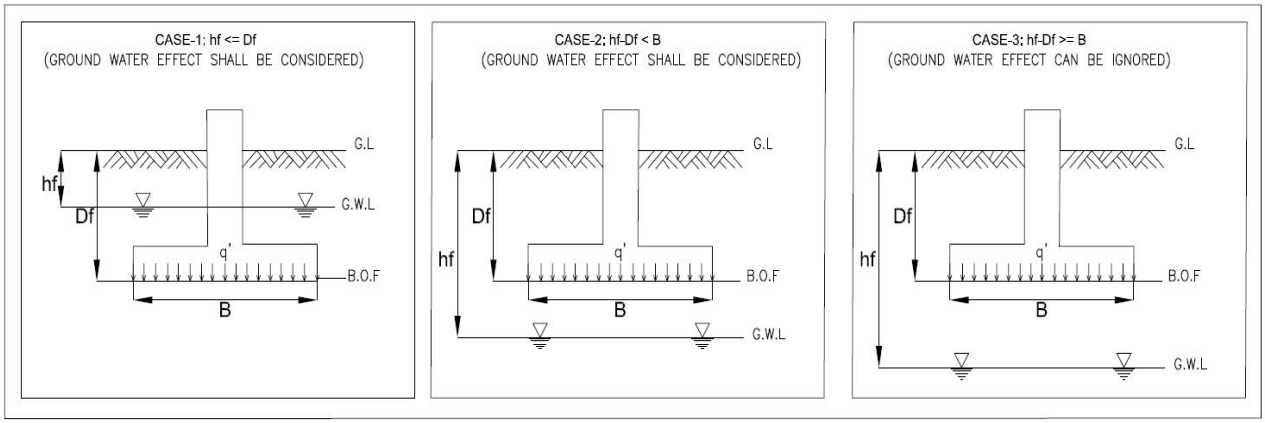
■Soil Parameter

Φ	c	Unit Wt(Above Base of Fdn.) γ_1	Unit Wt(Below Base of Fdn.) γ_2	Unit Wt of Water γ_w	G.W.L (h_f)
degree	kPa	kN/m^3	kN/m^3	kN/m^3	GL(-), m
34	0.00	17.00	17.00	10.00	5.00

■Foundation Dimension

Width, B	Eff. Width B'	Length, L	Eff. Length L'	Depth of Footing. D_f	Soil Slope	Fdn. Slope	Eff. Area, A_f
m	m	m	m	m	β, degree	η,degree	m^2
3	3.0	3	3	2	0	0	9.00

■Ground water cases: the Fig below shows different cases of ground water location.



- General Formula:

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma_2 \cdot B \cdot Nr \cdot Sr \cdot dr$

- The unit weights in the second and third term of the above equation shall be modified depending on the location of water table as explained in below.

- Case-1: hf<=Df

$q_u = c \cdot N_c \cdot S_c \cdot d_c + ((hf \cdot \gamma_1)+(Df-hf) \cdot (\gamma')) \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot (\gamma'_2) \cdot B \cdot Nr \cdot Sr \cdot dr \quad ; \quad \gamma'=\gamma-\gamma_w$

- Case-2: hf-Df<B

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot [(\gamma') + ((hf-Df)/B) \cdot (\gamma-\gamma'))] \cdot B \cdot Nr \cdot Sr \cdot dr$

- Case-3: hf-Df>=B

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma 2 \cdot B \cdot Nr \cdot Sr \cdot dr$

∴ Ground water case is: CASE-3

(B) Meyerhof's Method of Bearing Capacity Calculation

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma_2 \cdot B \cdot Nr \cdot Sr \cdot dr$

① Nc, Nq, Nr : Bearing capacity factor

Nc = 42.90 ; Nr = 32.82 ; Nq = 30.32

② Shape factor

Sc = 1.71 ; Sr = 1.35 ; Sq = 1.35 Kp = 3.54

Kp = TAN((45+Φ/2)*PI()/180)^2

Sc = 1+0.2*Kp*(B/L)

Sr = 1+0.1*Kp*(B/L)

Sq = Sr

③ **Depth factor**

dc = 1.25 ; dr = 1.13 ; dq = 1.13

dc = 1+0.2*(Kp^0.5)*(Df/L)

dr = 1+0.1*(Kp^0.5)*(Df/L)

dq = dr

■ Meyrhof's Bearing Capacity Factors

Angle	Nc	Nr	Nq
0	5.14	0	1
5	6.5	0.1	1.6
10	8.3	0.4	2.5
15	11	1.1	3.9
20	14.8	2.9	6.4
25	20.7	6.8	10.7
30	30.1	15.7	18.4
35	46.1	37.1	33.3
40	75.3	93.7	64.2
45	133.9	262.7	134.9
50	266.9	873.7	319

■ Effective Unit Weights

Item	Cases	Applied Formula	Value	Use
q'=	Case-1	$hf*\gamma_1 + (Df-hf)*\gamma'_1$	-	34
	Case-2 & 3	$Df*\gamma_1$	34	
γ'_2=	Case-1	$\gamma_2-\gamma_w$	-	17.0
	Case-2	$((\gamma_2-\gamma_w) +((hf-Df)/B) \cdot (\gamma_2-(\gamma_2-\gamma_w)))$	-	
	Case-3	γ_2	17	

$q_u = c \cdot N_c \cdot S_c \cdot d_c + q' \cdot Nq \cdot Sq \cdot dq + 0.5 \cdot \gamma_2 \cdot B \cdot Nr \cdot Sr \cdot dr$

$= 0.00 + 1570.49 + 1274.99$

$= 2845.47$ kPa

$q_a = 2845.47 / 3$

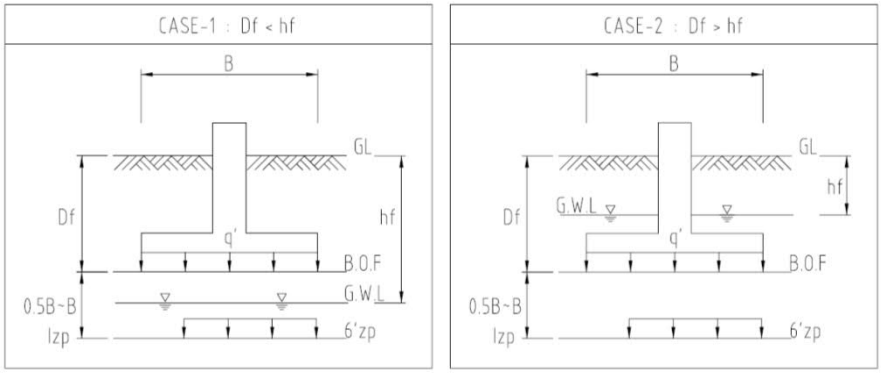
$q_a =$ 948.49 (kPa) ; is the the Allowabel Bearing Capacity (kPa)

2. Settlement Calculation For BH-50

■ Footing dimension

Width (B)	Length (L)	Depth (D_f)	h_f	L/B	Footing Shape	Unit Wt of Soil
3.00 m	3.00 m	2.00 m	5.00 m	1	SQU	17 kN/m3

■ Footing shape
L/B = 1, : Square (SQU) and Circular Footings (CIR)
L/B ≥ 10, : Continuous Footings (CON)
1< L/B < 10, : Rectangular Footings (REC)



■ Depth of influence = 8.0 m
where, Square Shape = Df+2B = 8.0 m
Continuous Shape = Df+4B = 14.0 m
* For Rec Shape it shall be determind By linear Interpolation of Square and Contineous Types -

■ Effective stress at a depth D_f below the ground surface (σ'vo)

- σ'vo = 34.00 kN/m^2

where, For Case-1 : $\sigma'_{vo} = (\gamma_s \times D_f)$
For Case-2 : $\sigma'_{vo} = (\gamma_s \times h_f) + (\gamma_s - \gamma_w) \times (D_f - h_f)$

■ Initial vertical effective stress at a depth of I_{zp} (σ'_{zp})

- σ'_{zp} = 59.50 kN/m^2

- $\sigma'_{zp(squ)}$ = 59.50 kN/m^2 : (For Square)

where, $\sigma'_{zp(squ)} = (\gamma_s \times h_f) + (\gamma_s - \gamma_w) \times (D_f + B/2 - h_f)$: For $h_f < D_f + B/2$
$\sigma'_{zp(squ)} = \gamma_s \times (D_f + B/2)$: For $h_f > D_f + B/2$

- $\sigma'_{zp(rec)}$ = - kN/m^2 : (Rec. Shape Footing)
* For Rec Shape it shall be determind By linear Interpolation of Square and Contineous Types

- $\sigma'_{zp(con)}$ = 85.00 kN/m^2 : (For Continuous Shape Footing)

where, $\sigma'_{zp(con)} = (\gamma_s \times h_f) + (\gamma_s - \gamma_w) \times (D_f + B - h_f)$: For $h_f < D_f + B$
$\sigma'_{zp(con)} = \gamma_s \times (D_f + B)$: For $h_f > D_f + B$

■ Bearing pressure at bottom of footing (q)

- q = 110.00 kN/m^2

■ Peak strain influence factor (I_{zp}) = 0.613

- $I_{zp(squ)} = 0.5 + 0.1 \sqrt{(q' / \sigma'_{zp(squ)})}$ = 0.61
- $I_{zp(con)} = 0.5 + 0.1 \sqrt{(q' / \sigma'_{zp(con)})}$ = 0.59
* For Rec Shape it shall be determind By linear Interpolation of Square and Contineous Types -

■ Depth factor (C_1)

- $C_1 = 1 - 0.5 (\sigma'_{vo} / q')$
  $= 0.776$

■ Secondary creep factor (C_2)

- $C_2 = 1 + 0.2 \log_{10} (t / 0.1)$
  $= 1.400$

■ Time since application of load (yr)

- $t =$ 10.0 year ($t \geq 0.1$ yr)

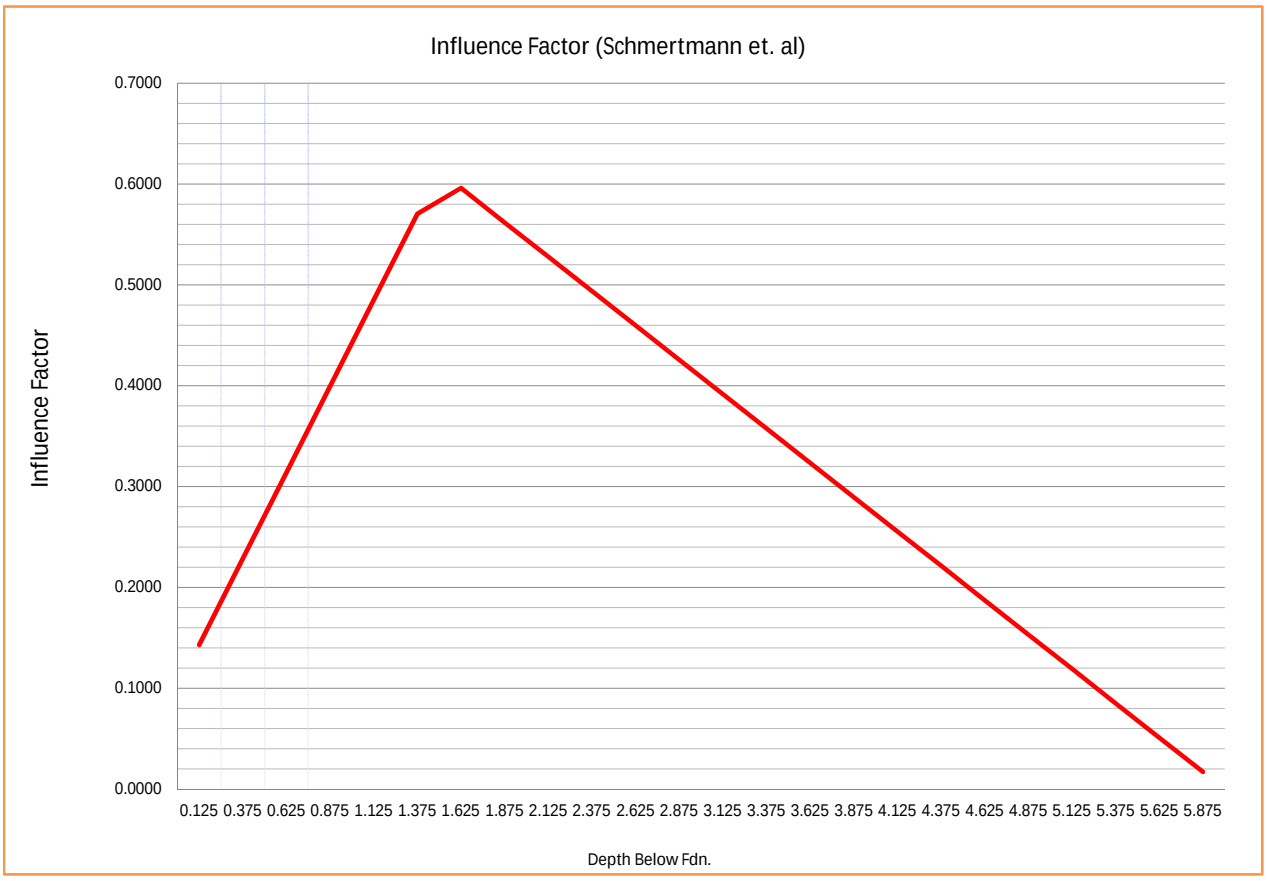
■ Shape factor (C_3)

- $C_3 = 1.03 - 0.03 (L / B) \geq 0.73$
  $= 1.000$

Depth interval = 0.25 m

	Top (m)	Bottom (m)	E_s (kN/m^2)	z_f (m)	$I_{z(squ)}$	$I_{z(rec)}$	$I_{z(con)}$	I_{zi}	H_i (m)	$I_{zi} H_i / E_s$	**δ** (mm)
	0.0	2.00									
	2.0	2.25	3,000	0.125	0.1428	NA	NA	0.1428	0.25	1.19E-05	0.9826
	2.3	2.50	3,000	0.375	0.2283	NA	NA	0.2283	0.25	1.90E-05	1.5712
	2.5	2.75	3,000	0.625	0.3138	NA	NA	0.3138	0.25	2.61E-05	2.1597
	2.8	3.00	3,000	0.875	0.3993	NA	NA	0.3993	0.25	3.33E-05	2.7482
	3.0	3.25	13,000	1.125	0.4848	NA	NA	0.4848	0.25	9.32E-06	0.7700
	3.3	3.50	13,000	1.375	0.5703	NA	NA	0.5703	0.25	1.10E-05	0.9058
	3.5	3.75	13,000	1.625	0.5960	NA	NA	0.5960	0.25	1.15E-05	0.9467
	3.8	4.00	22,000	1.875	0.5619	NA	NA	0.5619	0.25	6.39E-06	0.5275
	4.0	4.25	22,000	2.125	0.5279	NA	NA	0.5279	0.25	6.00E-06	0.4955
	4.3	4.50	22,000	2.375	0.4938	NA	NA	0.4938	0.25	5.61E-06	0.4635
	4.5	4.75	13,000	2.625	0.4598	NA	NA	0.4598	0.25	8.84E-06	0.7303
	4.8	5.00	13,000	2.875	0.4257	NA	NA	0.4257	0.25	8.19E-06	0.6762
	5.0	5.25	13,000	3.125	0.3917	NA	NA	0.3917	0.25	7.53E-06	0.6221
	5.3	5.50	2,100	3.375	0.3576	NA	NA	0.3576	0.25	4.26E-05	3.5163
	5.5	5.75	2,100	3.625	0.3235	NA	NA	0.3235	0.25	3.85E-05	3.1815
	5.8	6.00	2,100	3.875	0.2895	NA	NA	0.2895	0.25	3.45E-05	2.8466
	6.0	6.25	19,000	4.125	0.2554	NA	NA	0.2554	0.25	3.36E-06	0.2776
	6.3	6.50	19,000	4.375	0.2214	NA	NA	0.2214	0.25	2.91E-06	0.2406
	6.5	6.75	19,000	4.625	0.1873	NA	NA	0.1873	0.25	2.46E-06	0.2036
	6.8	7.00	19,000	4.875	0.1533	NA	NA	0.1533	0.25	2.02E-06	0.1666
	7.0	7.25	19,000	5.125	0.1192	NA	NA	0.1192	0.25	1.57E-06	0.1295
	7.3	7.50	19,000	5.375	0.0851	NA	NA	0.0851	0.25	1.12E-06	0.0925
	7.5	7.75	32,000	5.625	0.0511	NA	NA	0.0511	0.25	3.99E-07	0.0330
-End of Influence-	7.8	8.00	32,000	5.875	0.0170	NA	NA	0.0170	0.25	1.33E-07	0.0110

Σtotal **δ*** = 24 mm


Influence Factor (Schmertmann et. al)
Influence Factor
0.7000
0.6000
0.5000
0.4000
0.3000
0.2000
0.1000
0.0000
0.125 0.375 0.625 0.875 1.125 1.375 1.625 1.875 2.125 2.375 2.625 2.875 3.125 3.375 3.625 3.875 4.125 4.375 4.625 4.875 5.125 5.375 5.625 5.875
Depth Below Fdn.

II) Sample Vertical Pile Capacity Calculation sheet

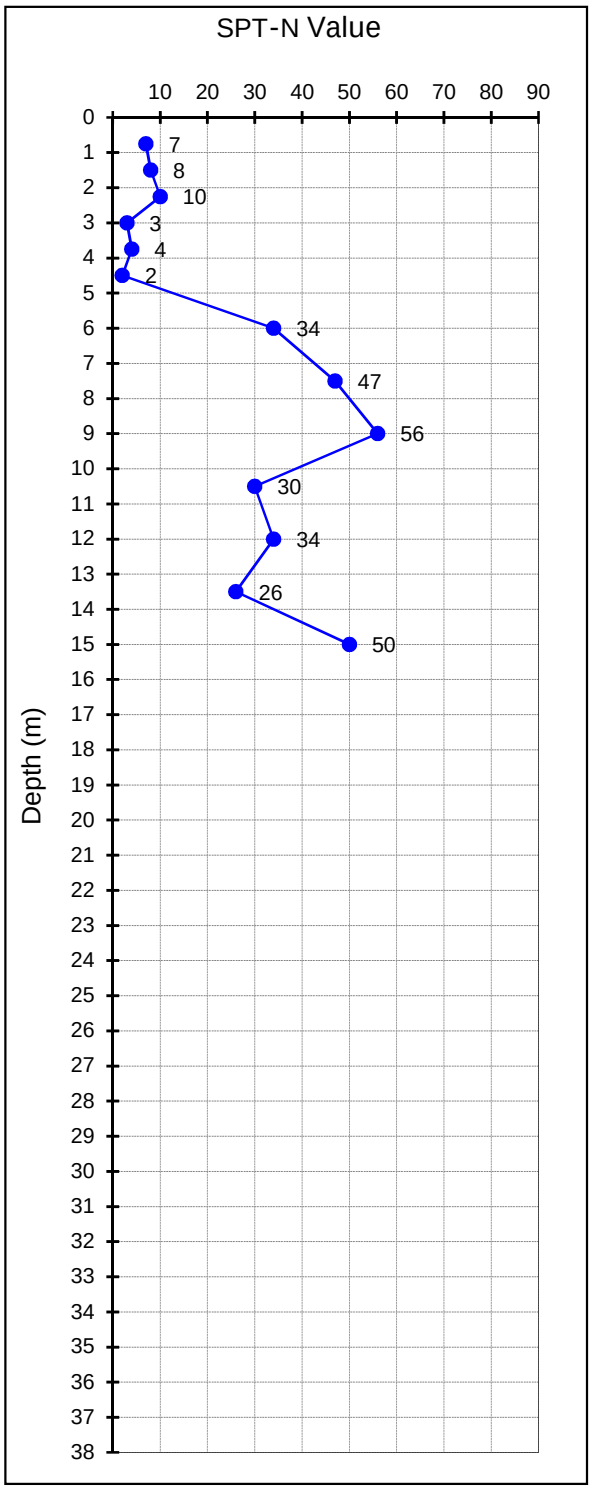
APPENDIX-1 : ALLOWABLE PILE BEARING CAPACITY

A1.1 GEOTHECHNICAL DATA

1) Standard Penetration Test Data - Tank Area(D800)

Hole No.	Planned EL.	Primarily EL.	Ground water level	Remark
BH - 07	EL.+7.00m	EL.+6.00m	EL.+2.00m	+1.00m Filling

Stratigraphy of Bore-hole				
No.	EL.+ m	Depth m	N-Value	Material
1	5.25	0.75	7	sand
2	4.50	1.50	8	sand
3	3.75	2.25	10	sand
4	3.00	3.00	3	sand
5	2.25	3.75	4	sand
6	1.50	4.50	2	sand
7	0.00	6.00	34	sand
8	-1.50	7.50	47	sand
9	-3.00	9.00	56	sand
10	-4.50	10.50	30	sand
11	-6.00	12.00	34	sand
12	-7.50	13.50	26	sand
13	-9.00	15.00	50	sand
14	-10.50	16.50	88	sand
15	-12.00	18.00	88	sand
16	-13.50	19.50	88	sand
17	-15.00	21.00	88	sand



2) Revised geological data

Layer No.	EL.+ (m)		Depth	N	Material	Remark
	from	to	m	blow		
L1	7.00	6.00	1.00	7	sand	FDN DEPTH
L2	6.00	5.25	0.75	7	sand	
L3	5.25	4.50	0.75	8	sand	
L4	4.50	3.75	0.75	10	sand	
L5	3.75	3.00	0.75	3	sand	
L6	3.00	2.25	0.75	4	sand	
L7	2.25	2.00	0.25	3	sand	Ground water level : EL.+2.00
L8	2.00	1.50	0.50	2	sand	
L9	1.50	0.00	1.50	34	sand	
L10	0.00	-1.50	1.50	47	sand	
L11	-1.50	-3.00	1.50	56	sand	
L12	-3.00	-4.50	1.50	30	sand	
L13	-4.50	-6.00	1.50	34	sand	
L14	-6.00	-7.50	1.50	26	sand	
L15	-7.50	-9.00	1.50	50	sand	
L16	-9.00	-10.50	1.50	88	sand	
L17	-10.50	-12.00	1.50	88	sand	
L18	-12.00	-13.50	1.50	88	sand	
L19	-13.50	-15.00	1.50	88	sand	

A1.2 PILE INFORMATION

1) Pile Data

- Pile Diameter (D) = 800 mm
- Pile Area (A) = 0.50 m^2
- Foundation Bottom Level (Df)
  = 1.00 m

A1.3 CALCULATION OF DESIGN SUMMARY

1) Allowable Vertical Pile Bearing Capacity

Case	Length (m)	Allowable Capacity(kN/EA)	Allowable Capacity(kN/EA) (Applied to AASHTO 7th)
By AASHTO 7th	12.0	1947.35	**1947.35**

2) Allowable Uplift Pile Bearing Capacity

Case	Length (m)	Allowable Capacity(kN/EA)	Allowable Capacity(kN/EA) (Applied to AASHTO 7th)
By AASHTO 7th	12.0	1060.26	**1060.26**

A1.4 ALLOWABLE VERTICAL PILE BEARING CAPACITY

1) By AASHTO 7th

Where , A_p : Pile Area
= 0.50 m^2
FS : Safety Factor = 2

Stratigraphy of Bore-hole			(N1)60	φ'f (°)	σp' (kPa)	β	Side Resistance (kN)	Accu. Side Resistance (kN)	Tip Resistance (kN)	Ultimate Comp. Capacity (kN)	Allowable Comp. Capacity (kN)
No.	Depth m	N-Value									
1	0.00	-									
2	0.75	7	11.6	37	4.73	1.66	42.15	42.15	202.25	244.40	122.20
3	1.50	8	11.403	37	5.26	1.16	58.98	101.13	231.14	332.27	166.14
4	2.25	10	12.897	38	6.29	1.02	78.21	179.34	288.93	468.27	234.13
5	3.00	3	3.5805	33	2.40	0.44	45.10	224.44	86.68	311.12	155.56
6	3.75	4	4.4756	33	3.02	0.45	57.46	281.90	115.57	397.47	198.74
7	4.00	3	3.2919	32	2.40	0.38	17.10	299.00	86.68	385.68	192.84
8	4.50	2	2.1158	30	1.73	0.29	29.90	328.90	57.79	386.69	193.34
9	6.00	34	34.398	42	16.73	1.25	437.57	766.47	982.35	1748.82	874.41
10	7.50	47	45.643	43	21.68	1.41	557.06	1323.53	1357.95	2681.49	1340.74
11	9.00	56	52.357	43	24.95	1.46	642.90	1966.44	1444.63	3411.07	1705.53
12	10.50	30	27.069	41	15.14	0.92	448.49	2414.92	866.78	3281.70	1640.85
13	12.00	34	29.667	41	16.73	0.94	497.43	2912.35	982.35	3894.70	1947.35
14	13.50	26	21.976	40	13.50	0.76	435.53	3347.88	751.21	4099.09	2049.54
15	15.00	50	40.999	42	22.78	1.06	660.48	4008.36	1444.63	5452.99	2726.49
16	16.50	88	70.093	44	35.81	1.44	963.17	4971.53	1444.63	6416.16	3208.08
17	18.00	88	68.162	44	35.81	1.38	979.99	5951.52	1444.63	7396.15	3698.08
18	19.50	88	66.351	44	35.81	1.31	996.13	6947.65	1444.63	8392.28	4196.14
19	21.00	88	64.644	44	35.81	1.26	1011.65	7959.30	1444.63	9403.93	4701.96
Use Qa											1947.35

2) Allowable Vertical Pile Bearing Capacity

Method	Allowable Vertical Pile Bearing Capacity (kN/EA)
By AASHTO 7th	1947.35

A1.5 ALLOWABLE UPLIFT PILE BEARING CAPACITY

1) By AASHTO 7th

Where , FS : Safety Factor = 3

Stratigraphy of Bore-hole			(N1)60	φ'f (°)	σp' (kPa)	β	Side Resistance (kN)	Accu. Side Resistance (kN)	Tip Resistance (kN)	Pile Weight (W)	Allowable Uplift Capacity (kN)
No.	Depth m	N-Value									
1	0.00	-									
2	0.75	7	11.6	37	4.73	1.66	42.15	42.15	202.25	10.30	24.36
3	1.50	8	11.403	37	5.26	1.16	58.98	101.13	231.14	15.58	49.29
4	2.25	10	12.897	38	6.29	1.02	78.21	179.34	288.93	20.86	80.64
5	3.00	3	3.5805	33	2.40	0.44	45.10	224.44	86.68	26.14	100.95
6	3.75	4	4.4756	33	3.02	0.45	57.46	281.90	115.57	31.42	125.38
7	4.00	3	3.2919	32	2.40	0.38	17.10	299.00	86.68	33.18	132.84
8	4.50	2	2.1158	30	1.73	0.29	29.90	328.90	57.79	36.69	146.33
9	6.00	34	34.398	42	16.73	1.25	437.57	766.47	982.35	47.25	302.74
10	7.50	47	45.643	43	21.68	1.41	557.06	1323.53	1357.95	57.81	498.98
11	9.00	56	52.357	43	24.95	1.46	642.90	1966.44	1444.63	68.36	723.84
12	10.50	30	27.069	41	15.14	0.92	448.49	2414.92	866.78	78.92	883.89
13	12.00	34	29.667	41	16.73	0.94	497.43	2912.35	982.35	89.47	1060.26
14	13.50	26	21.976	40	13.50	0.76	435.53	3347.88	751.21	100.03	1215.99
15	15.00	50	40.999	42	22.78	1.06	660.48	4008.36	1444.63	110.58	1446.70
16	16.50	88	70.093	44	35.81	1.44	963.17	4971.53	1444.63	121.14	1778.32
17	18.00	88	68.162	44	35.81	1.38	979.99	5951.52	1444.63	131.70	2115.54
18	19.50	88	66.351	44	35.81	1.31	996.13	6947.65	1444.63	142.25	2458.13
19	21.00	88	64.644	44	35.81	1.26	1011.65	7959.30	1444.63	152.81	2805.91
Use Qa											1060.26

2) Allowable Uplift Pile Bearing Capacity

Method	Allowable Uplift Pile Bearing Capacity (kN/EA)
By AASHTO 7th	1060.26

Table 0. Pile Ground Investigation Coefficient

Ground	SPT N Value	Unit Weight	Angle of internal friction
Sabkha	0-5	17	27
Loose Sand	5-10	17.5	31
Medium dense Sand	10-30	18	34
Dense Sand	30-50	18.5	36
Very dense Sand	≤50	19	39

Table 1. Bearing Capacity Factor Nq

Φ(°)	26	28	30	31	32	33	34	35	36	37	38	39	40
Nq (Drived Piles)	10	15	21	24	29	35	42	50	62	77	89	120	145
Nq (Drilled Piles)	5	8	10	12	14	17	21	25	30	38	43	60	72

Table 2. Eqrth Pressure Coefficient K

Pile Type	Ks
Driven single displacement piles	1.0~1.5
Drilled Piles	0.7

Table 3. Friction Angle δ

Pile Type	δ
Steel	20˚
Concrete	(3/4) x Φ
Timber	(3/4) x Φ

Table 4. Bearing Capacity Factor, Nc

Pile Toe Diameter (m)	Nc
<0.5	9
0.50-1.0	7
>1.0	6

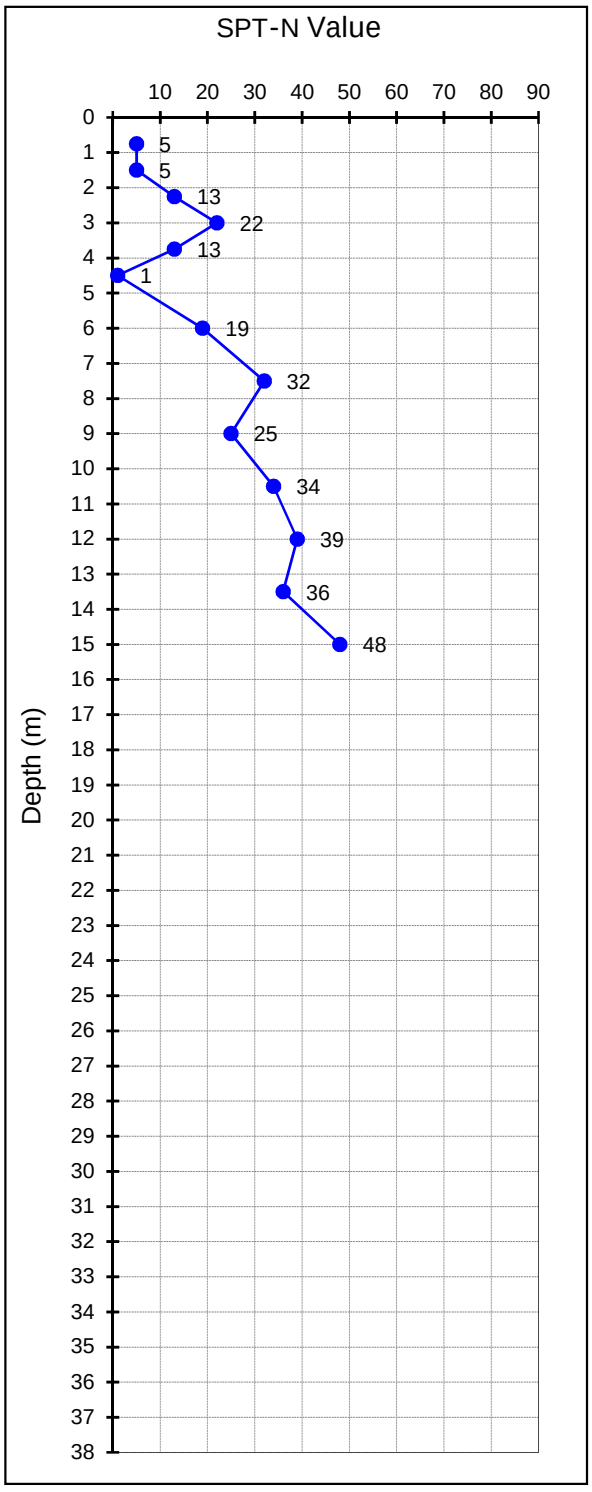
APPENDIX-1 : ALLOWABLE PILE BEARING CAPACITY

A1.1 GEOTHECHNICAL DATA

1) Standard Penetration Test Data - Utility Area(Tank)

Hole No.	Planned EL.	Primarily EL.	Ground water level	Remark
BH - 50	EL.+7.00m	EL.+6.00m	EL.+2.00m	+1.00m Filling

Stratigraphy of Bore-hole				
No.	EL.+ m	Depth m	N-Value	Material
1	5.25	0.75	5	sand
2	4.50	1.50	5	sand
3	3.75	2.25	13	sand
4	3.00	3.00	22	sand
5	2.25	3.75	13	sand
6	1.50	4.50	1	sand
7	0.00	6.00	19	sand
8	-1.50	7.50	32	sand
9	-3.00	9.00	25	sand
10	-4.50	10.50	34	sand
11	-6.00	12.00	39	sand
12	-7.50	13.50	36	sand
13	-9.00	15.00	48	sand
14	-10.50	16.50	50	sand
15	-12.00	18.00	50	sand
16	-13.50	19.50	50	sand
17	-15.00	21.00	50	sand



2) Revised geological data

Layer No.	EL.+ (m)		Depth	N	Material	Remark
	from	to	m	blow		
L1	7.00	6.00	1.00	5	sand	FDN DEPTH
L2	6.00	5.25	0.75	5	sand	
L3	5.25	4.50	0.75	5	sand	
L4	4.50	3.75	0.75	13	sand	
L5	3.75	3.00	0.75	22	sand	
L6	3.00	2.25	0.75	13	sand	
L7	2.25	2.00	0.25	8	sand	Ground water level : EL.+2.00
L8	2.00	1.50	0.50	1	sand	
L9	1.50	0.00	1.50	19	sand	
L10	0.00	-1.50	1.50	32	sand	
L11	-1.50	-3.00	1.50	25	sand	
L12	-3.00	-4.50	1.50	34	sand	
L13	-4.50	-6.00	1.50	39	sand	
L14	-6.00	-7.50	1.50	36	sand	
L15	-7.50	-9.00	1.50	48	sand	
L16	-9.00	-10.50	1.50	50	sand	
L17	-10.50	-12.00	1.50	50	sand	
L18	-12.00	-13.50	1.50	50	sand	
L19	-13.50	-15.00	1.50	50	sand	

A1.2 PILE INFORMATION

1) Pile Data

- Pile Diameter (D) = 600 mm
- Pile Area (A) = 0.28 m^2
- Foundation Bottom Level (Df)
  = 1.00 m

A1.3 CALCULATION OF DESIGN SUMMARY

1) Allowable Vertical Pile Bearing Capacity

Case	Length (m)	Allowable Capacity(kN/EA)	Allowable Capacity(kN/EA) (Applied to AASHTO 7th)
By AASHTO 7th	12.0	1302.90	**1302.90**

2) Allowable Uplift Pile Bearing Capacity

Case	Length (m)	Allowable Capacity(kN/EA)	Allowable Capacity(kN/EA) (Applied to AASHTO 7th)
By AASHTO 7th	12.0	707.65	**707.65**

A1.4 ALLOWABLE VERTICAL PILE BEARING CAPACITY

1) By AASHTO 7th

Where , A_p : Pile Area
= 0.28 m^2
FS : Safety Factor = 2

Stratigraphy of Bore-hole			(N1)60	φ'f (°)	σp' (kPa)	β	Side Resistance (kN)	Accu. Side Resistance (kN)	Tip Resistance (kN)	Ultimate Comp. Capacity (kN)	Allowable Comp. Capacity (kN)
No.	Depth m	N-Value									
1	0.00	-									
2	0.75	5	8.2857	36	3.61	1.34	25.54	25.54	81.26	106.80	53.40
3	1.50	5	7.1266	35	3.61	0.88	33.41	58.95	81.26	140.21	70.11
4	2.25	13	16.766	39	7.76	1.20	68.85	127.80	211.28	339.07	169.54
5	3.00	22	26.257	41	11.81	1.38	105.28	233.08	357.55	590.63	295.31
6	3.75	13	14.546	38	7.76	0.86	82.26	315.34	211.28	526.61	263.31
7	4.00	8	8.7784	36	5.26	0.63	21.30	336.64	130.02	466.65	233.33
8	4.50	1	1.0579	28	1.00	0.22	16.77	353.41	16.25	369.66	184.83
9	6.00	19	19.223	39	10.51	0.87	229.96	583.37	308.79	892.16	446.08
10	7.50	32	31.076	41	15.94	1.10	327.98	911.35	520.07	1431.42	715.71
11	9.00	25	23.374	40	13.09	0.88	292.20	1203.55	406.30	1609.85	804.93
12	10.50	34	30.678	41	16.73	0.99	362.86	1566.41	552.57	2118.98	1059.49
13	12.00	39	34.029	42	18.68	1.02	405.56	1971.97	633.83	2605.80	1302.90
14	13.50	36	30.428	41	17.52	0.92	396.15	2368.12	585.08	2953.20	1476.60
15	15.00	48	39.359	42	22.05	1.04	483.03	2851.16	780.10	3631.26	1815.63
16	16.50	50	39.825	42	22.78	1.01	505.72	3356.88	812.60	4169.48	2084.74
17	18.00	50	38.729	42	22.78	0.96	515.66	3872.54	812.60	4685.14	2342.57
18	19.50	50	37.699	42	22.78	0.92	525.22	4397.76	812.60	5210.36	2605.18
19	21.00	50	36.73	42	22.78	0.89	534.44	4932.20	812.60	5744.80	2872.40
Use Qa											1302.90

2) Allowable Vertical Pile Bearing Capacity

Method	Allowable Vertical Pile Bearing Capacity (kN/EA)
By AASHTO 7th	1302.90

A1.5 ALLOWABLE UPLIFT PILE BEARING CAPACITY

1) By AASHTO 7th

Where , FS : Safety Factor = 3

Stratigraphy of Bore-hole			(N1)60	φ'f (°)	σp' (kPa)	β	Side Resistance (kN)	Accu. Side Resistance (kN)	Tip Resistance (kN)	Pile Weight (W)	Allowable Uplift Capacity (kN)
No.	Depth m	N-Value									
1	0.00	-									
2	0.75	5	8.2857	36	3.61	1.34	25.54	25.54	81.26	5.80	14.31
3	1.50	5	7.1266	35	3.61	0.88	33.41	58.95	81.26	8.77	28.42
4	2.25	13	16.766	39	7.76	1.20	68.85	127.80	211.28	11.73	54.33
5	3.00	22	26.257	41	11.81	1.38	105.28	233.08	357.55	14.70	92.40
6	3.75	13	14.546	38	7.76	0.86	82.26	315.34	211.28	17.67	122.78
7	4.00	8	8.7784	36	5.26	0.63	21.30	336.64	130.02	18.66	130.87
8	4.50	1	1.0579	28	1.00	0.22	16.77	353.41	16.25	20.64	138.44
9	6.00	19	19.223	39	10.51	0.87	229.96	583.37	308.79	26.58	221.03
10	7.50	32	31.076	41	15.94	1.10	327.98	911.35	520.07	32.52	336.30
11	9.00	25	23.374	40	13.09	0.88	292.20	1203.55	406.30	38.45	439.64
12	10.50	34	30.678	41	16.73	0.99	362.86	1566.41	552.57	44.39	566.53
13	12.00	39	34.029	42	18.68	1.02	405.56	1971.97	633.83	50.33	707.65
14	13.50	36	30.428	41	17.52	0.92	396.15	2368.12	585.08	56.27	845.64
15	15.00	48	39.359	42	22.05	1.04	483.03	2851.16	780.10	62.20	1012.59
16	16.50	50	39.825	42	22.78	1.01	505.72	3356.88	812.60	68.14	1187.10
17	18.00	50	38.729	42	22.78	0.96	515.66	3872.54	812.60	74.08	1364.93
18	19.50	50	37.699	42	22.78	0.92	525.22	4397.76	812.60	80.02	1545.94
19	21.00	50	36.73	42	22.78	0.89	534.44	4932.20	812.60	85.95	1730.02
Use Qa											707.65

2) Allowable Uplift Pile Bearing Capacity

Method	Allowable Uplift Pile Bearing Capacity (kN/EA)
By AASHTO 7th	707.65

Table 0. Pile Ground Investigation Coefficient

Ground	SPT N Value	Unit Weight	Angle of internal friction
Sabkha	0-5	17	27
Loose Sand	5-10	17.5	31
Medium dense Sand	10-30	18	34
Dense Sand	30-50	18.5	36
Very dense Sand	≤50	19	39

Table 1. Bearing Capacity Factor Nq

Φ(°)	26	28	30	31	32	33	34	35	36	37	38	39	40
Nq (Drived Piles)	10	15	21	24	29	35	42	50	62	77	89	120	145
Nq (Drilled Piles)	5	8	10	12	14	17	21	25	30	38	43	60	72

Table 2. Eqrth Pressure Coefficient K

Pile Type	Ks
Driven single displacement piles	1.0~1.5
Drilled Piles	0.7

Table 3. Friction Angle δ

Pile Type	δ
Steel	20˚
Concrete	(3/4) x Φ
Timber	(3/4) x Φ

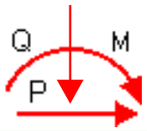
Table 4. Bearing Capacity Factor, Nc

Pile Toe Diameter (m)	Nc
<0.5	9
0.50-1.0	7
>1.0	6

III) Sample lateral Pile Capacity Calculation Sheet

FOUNDATION PROFILE & SOIL CONDITIONS

Diameter more than 24in (61cm).
For bell section, select "Belled" in Diameter Variation of Pile Section Screen. Recommendation: 2 to 4 in Item 3 of Page F.



FOUNDATION PROPERTIES

Depth	Width-cm	A'-cm2	Per.-cm	I'-cm4	E -MPa	W -kN/m
0.0	60	3082.5	188.5	637274.4	25743	6.984
Concrete (rough)						
12.0						

SOIL PROPERTIES

Depth	γ -kN/m3	ϕ	C-kN/m2	k-MN/m3	e50 %	Nspt
0.0	17	27	0.0	3.9		5
Sand/Gravel						
2.3	18	34	0.0	23.3		16
Sand/Gravel						
4.0	8	34	0.0	15.0		16
Sand/Gravel						
4.5	8	34	0.0	15.7		17
Sand/Gravel						
9.0	8.5	36	0.0	27.3		32
Sand/Gravel						
13.5	8.5	36	0.0	36.5		44
Sand/Gravel						

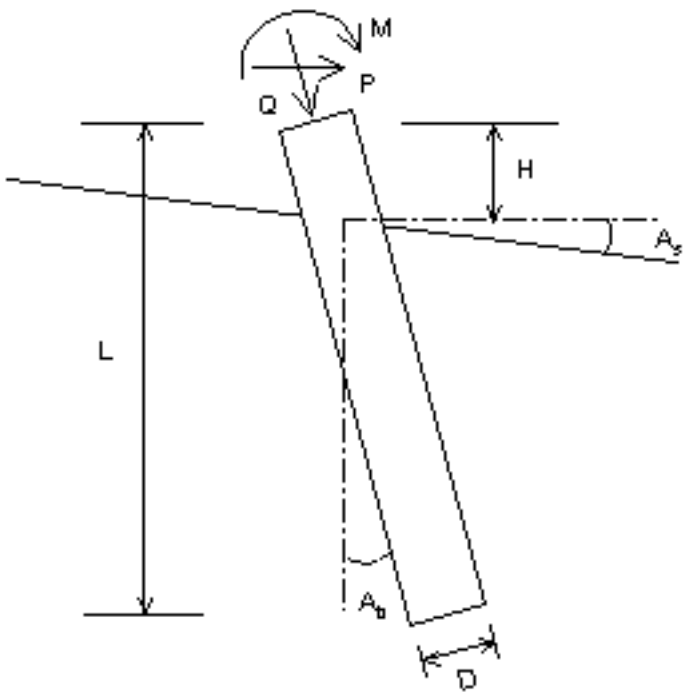
Depth from Ground-m: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Elevation from Ground-m: 6.0 5.0 4.0 3.0 2.0 1.0 0.0 -1.0 -2.0 -3.0 -4.0 -5.0 -6.0 -7.0 -8.0 -9.0

Batter Angle=0 (Pile diameter not to scale) Surface Angle=0

LATERAL ANALYSIS

Figure 2



Drilled Shaft (dia >24 in. or 61 cm)

Loads:

Load Factor for Vertical Loads= 1.0
Load Factor for Lateral Loads= 1.0
Loads Supported by Pile Cap= 0 %
Shear Condition: Static

(with Load Factor)
Vertical Load, Q= 1500.0 -kN
Shear Load, P= 110.0 -kN
Slope Restrain St= 0.00000 -cm/-cm

Profile:

Pile Length, L= 12.0 -m
Top Height, H= 0 -m
Slope Angle, As= 0
Batter Angle, Ab= 0
Fixed Head Condition

Soil Data:

Depth -m	Gamma -kN/m3	Phi	C -kN/m2	K -MN/m3	e50 or Dr %	Nspt
0	17	27	0.0	3.9	19.62	5
2.25	18	34	0.0	23.3	48.08	16
4.0	8	34	0.0	15.0	48.08	16
4.5	8	34	0.0	15.7	49.45	17
9	8.5	36	0.0	27.3	67.66	32
13.5	8.5	36	0.0	36.5	79.08	44
18	8.5	36	0.0	42.8	85.83	50

Pile Data:

Depth -m	Width -cm	Area -cm2	Per. -cm	I -cm4	E -MPa	Weight -kN/m
0.0	60	3082.5	188.5	637274.4	25743	6.984
12.0						

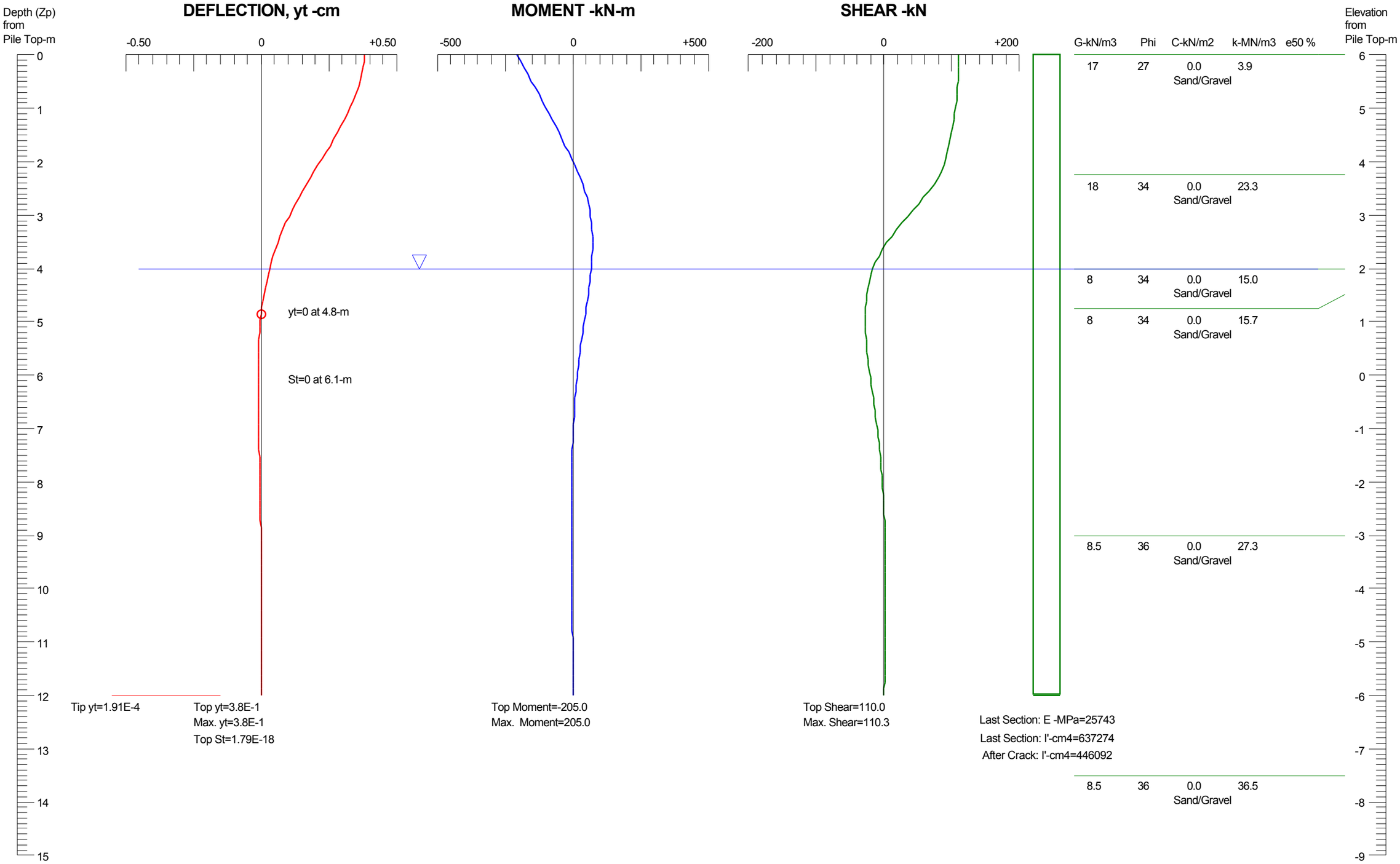
Single Pile Lateral Analysis:

Top Deflection, yt= 0.38000-cm
Max. Moment, M= -205.00-kN-m
Top Deflection Slope, St= 0.00000
OK! Top Deflection, 0.3800-cm is less than the Allowable Deflection= 2.50-cm

Note: If the program cannot find a result or the result exceeds the upper limit. The result will be displayed as 99999.
The Max. Moment calculated by program is an internal force from the applied load conditions. Structural engineer has to check whether the pile has enough capacity to resist the moment with adequate factor of safety. If not, the pile may fail under the load conditions.

PILE DEFLECTION & FORCE vs DEPTH

Single Pile, Khead=5, Kbc=2



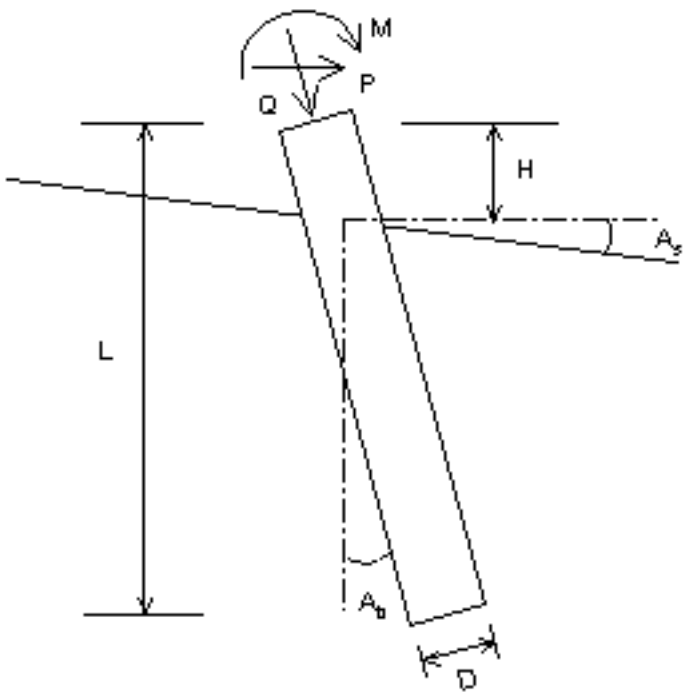
CivilTech Software

AMP-3
Dia. 600mm Pile Capa.

Figure 2

LATERAL ANALYSIS

Figure 2



Drilled Shaft (dia >24 in. or 61 cm)

Loads:

Load Factor for Vertical Loads= 1.0
Load Factor for Lateral Loads= 1.0
Loads Supported by Pile Cap= 0 %
Shear Condition: Static

(with Load Factor)
Vertical Load, Q= 1500.0 -kN
Shear Load, P= 110.0 -kN
Moment, M= 0.0 -kN-m

Profile:

Pile Length, L= 12.0 -m
Top Height, H= 0 -m
Slope Angle, As= 0
Batter Angle, Ab= 0

Soil Data:

Depth -m	Gamma -kN/m3	Phi	C -kN/m2	K -MN/m3	e50 or Dr %	Nspt
0	17	27	0.0	4.5	20.99	5
2.25	18	34	0.0	22.7	47.56	16
4.0	8	34	0.0	14.8	47.73	16
4.5	8	34	0.0	15.6	49.28	17
9	8.5	36	0.0	27.3	67.76	32
13.5	8.5	36	0.0	36.5	79.08	44
18	8.5	36.0	0.0	42.8	85.83	50

Pile Data:

Depth -m	Width -cm	Area -cm2	Per. -cm	I -cm4	E -MPa	Weight -kN/m
0.0	60	3082.5	188.5	637274.4	25743	6.984
12.0						

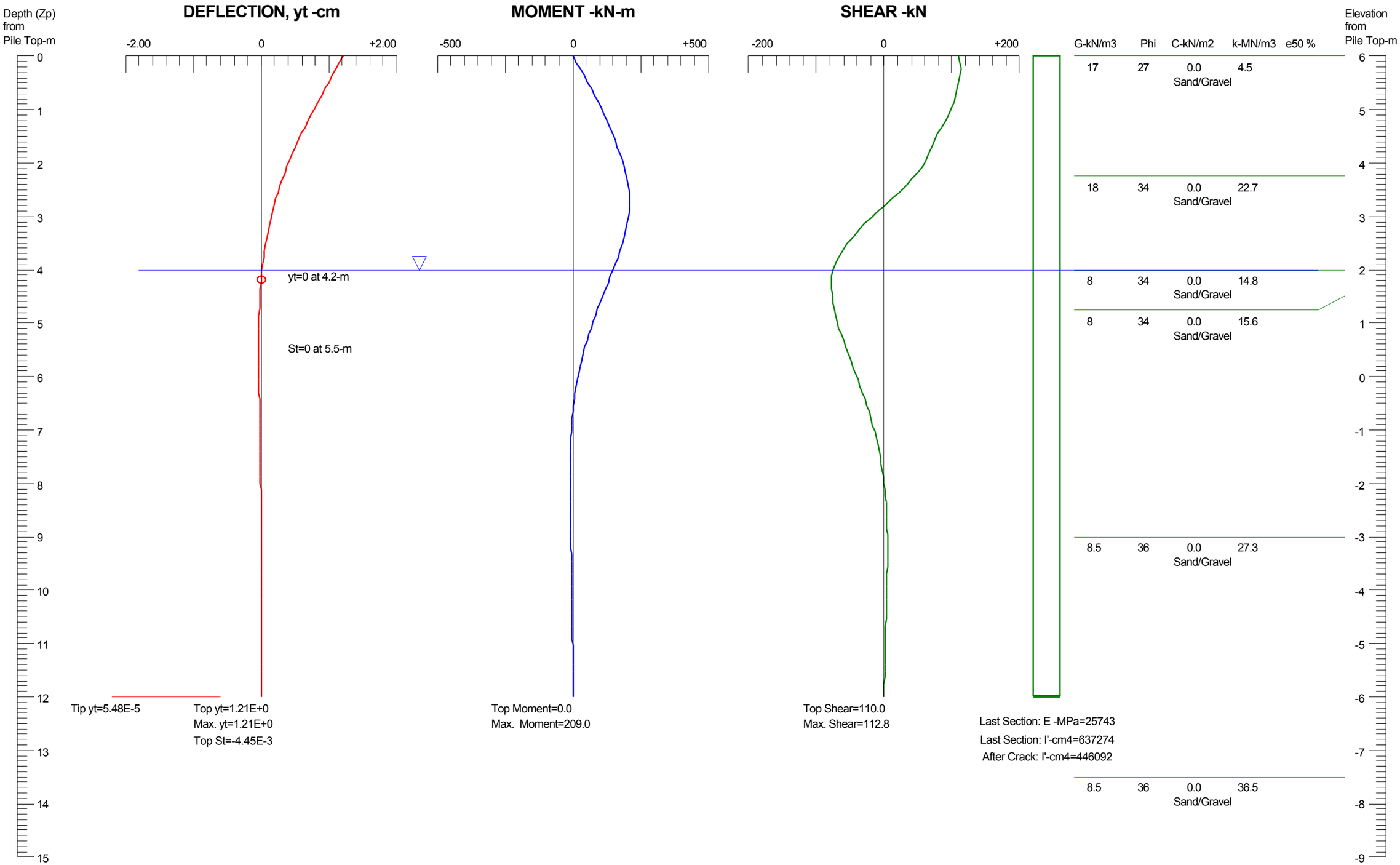
Single Pile Lateral Analysis:

Top Deflection, yt= 1.21000-cm
Max. Moment, M= 209.00-kN-m
Top Deflection Slope, St= -0.00445
OK! Top Deflection, 1.2100-cm is less than the Allowable Deflection= 2.50-cm

Note: If the program cannot find a result or the result exceeds the upper limit. The result will be displayed as 99999.
The Max. Moment calculated by program is an internal force from the applied load conditions. Structural engineer has to check whether the pile has enough capacity to resist the moment with adequate factor of safety. If not, the pile may fail under the load conditions.

PILE DEFLECTION & FORCE vs DEPTH

Single Pile, Khead=1, Kbc=1



CivilTech Software

AMP-3
Dia. 600mm Pile Capa.

Figure 2

FOUNDATION PROFILE & SOIL CONDITIONS

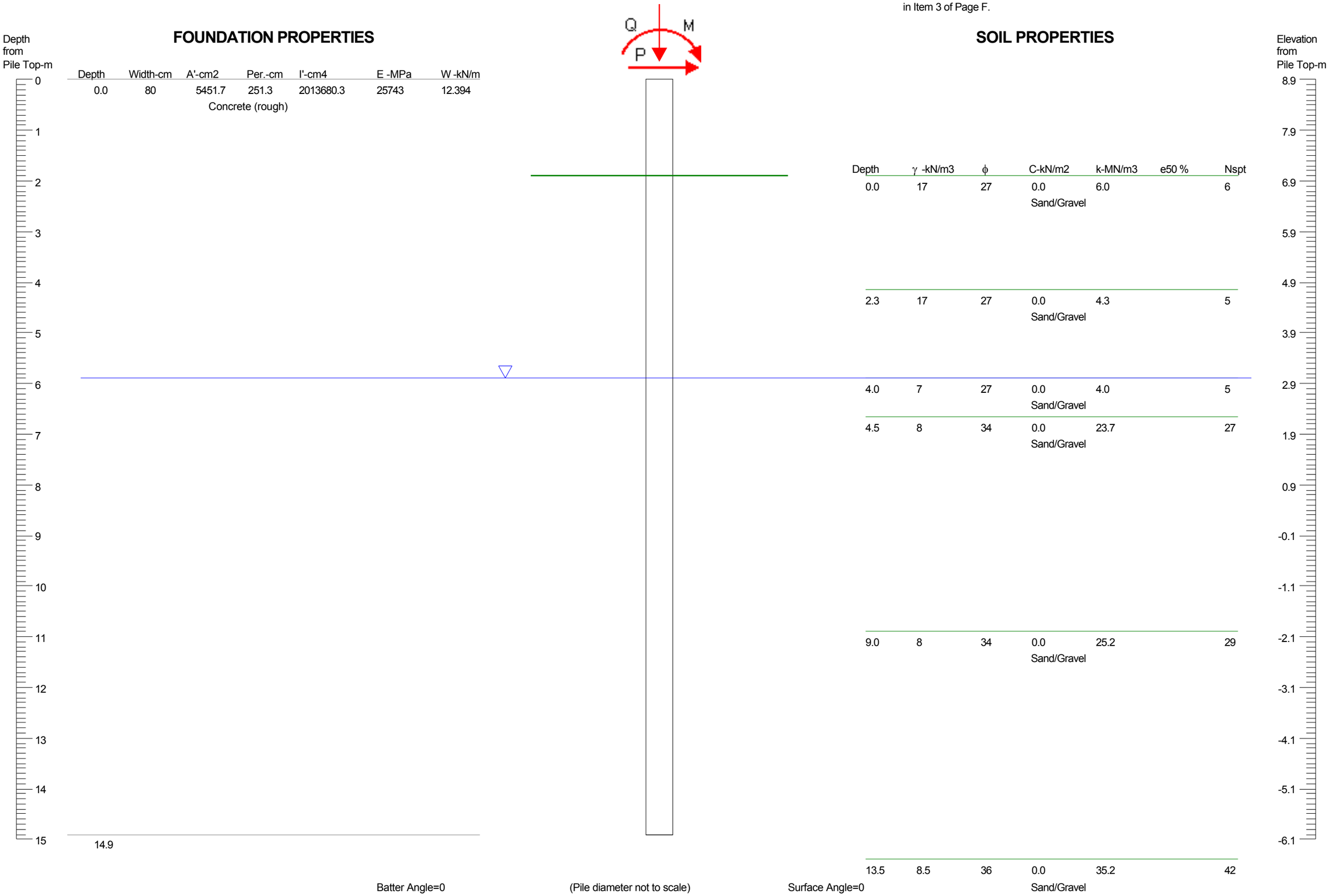
Diameter more than 24in (61cm).
For bell section, select "Belled" in Diameter Variation of Pile Section Screen. Recommendation: 2 to 4 in Item 3 of Page F.

FOUNDATION PROPERTIES

Depth	Width-cm	A'-cm2	Per.-cm	I'-cm4	E -MPa	W -kN/m
0.0	80	5451.7	251.3	2013680.3	25743	12.394
Concrete (rough)						

SOIL PROPERTIES

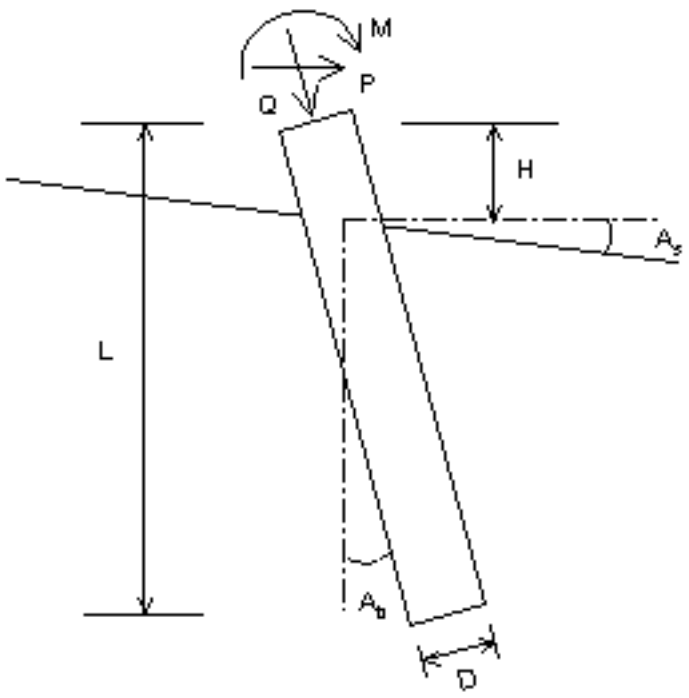
Depth	γ -kN/m3	ϕ	C-kN/m2	k-MN/m3	e50 %	Nspt
0.0	17	27	0.0	6.0		6
Sand/Gravel						
2.3	17	27	0.0	4.3		5
Sand/Gravel						
4.0	7	27	0.0	4.0		5
Sand/Gravel						
4.5	8	34	0.0	23.7		27
Sand/Gravel						
9.0	8	34	0.0	25.2		29
Sand/Gravel						
13.5	8.5	36	0.0	35.2		42
Sand/Gravel						



Batter Angle=0 (Pile diameter not to scale) Surface Angle=0

LATERAL ANALYSIS

Figure 2



Drilled Shaft (dia >24 in. or 61 cm)

Loads:

Load Factor for Vertical Loads= 1.0
Load Factor for Lateral Loads= 1.0
Loads Supported by Pile Cap= 0 %
Shear Condition: Static

(with Load Factor)
Vertical Load, Q= 2000.0 -kN
Shear Load, P= 130.0 -kN
Slope Restrain St= 0.00000 -cm/-cm

Profile:

Pile Length, L= 14.9 -m
Top Height, H= 1.9 -m
Slope Angle, As= 0
Batter Angle, Ab= 0
Fixed Head Condition

Soil Data:

Depth -m	Gamma -kN/m3	Phi	C -kN/m2	K -MN/m3	e50 or Dr %	Nspt
0	17	27	0.0	6.0	24.26	6
2.25	17	27	0.0	4.3	20.65	5
4.0	7	27	0.0	4.0	20.99	5
4.5	8	34	0.0	23.7	62.61	27
9	8	34	0.0	25.2	64.80	29
13.5	8.5	36	0.0	35.2	77.58	42
18	9	39	0.0	42.8	85.61	50

Pile Data:

Depth -m	Width -cm	Area -cm2	Per. -cm	I -cm4	E -MPa	Weight -kN/m
0.0	80	5451.7	251.3	2013680.3	25743	12.394
14.9						

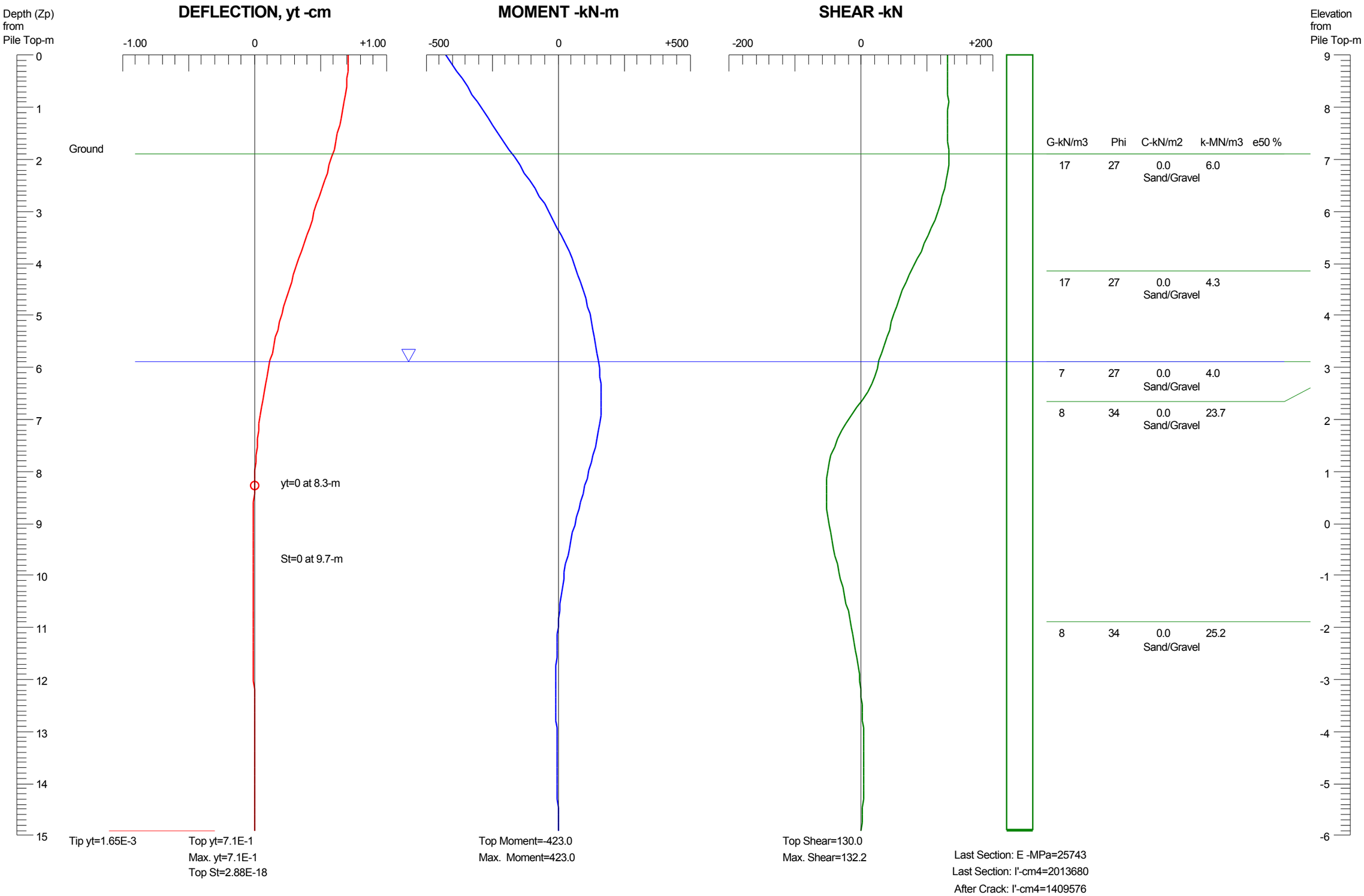
Single Pile Lateral Analysis:

Top Deflection, yt= 0.71000-cm
Max. Moment, M= -423.00-kN-m
Top Deflection Slope, St= 0.00000
OK! Top Deflection, 0.7100-cm is less than the Allowable Deflection= 2.50-cm

Note: If the program cannot find a result or the result exceeds the upper limit. The result will be displayed as 99999.
The Max. Moment calculated by program is an internal force from the applied load conditions. Structural engineer has to check whether the pile has enough capacity to resist the moment with adequate factor of safety. If not, the pile may fail under the load conditions.

PILE DEFLECTION & FORCE vs DEPTH

Single Pile, Khead=5, Kbc=2



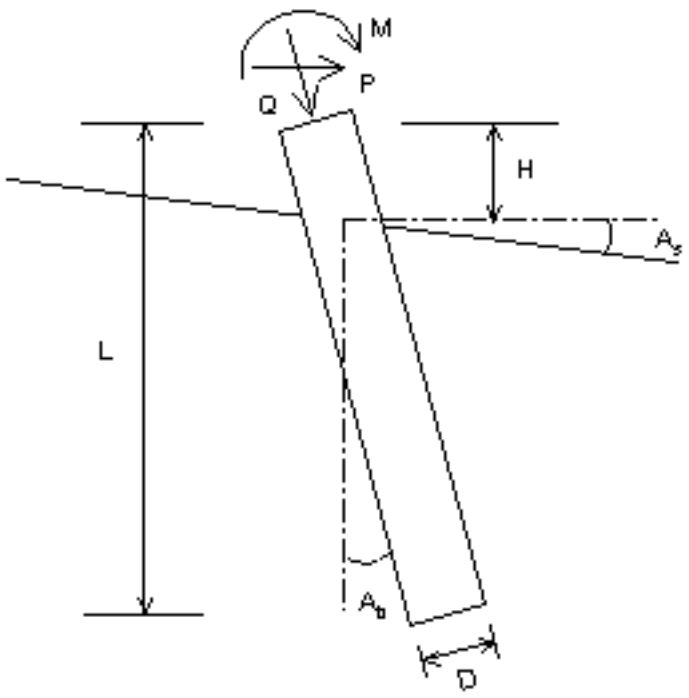
CivilTech Software

AMP-3_D800_BH-7
Fixed End (P=2000kN, V=130kN)

Figure 2

LATERAL ANALYSIS

Figure 2



Drilled Shaft (dia >24 in. or 61 cm)

Loads:

Load Factor for Vertical Loads= 1.0
Load Factor for Lateral Loads= 1.0
Loads Supported by Pile Cap= 0 %
Shear Condition: Static

(with Load Factor)
Vertical Load, Q= 2000.0 -kN
Shear Load, P= 130.0 -kN
Moment, M= 0.0 -kN-m

Profile:

Pile Length, L= 13.0 -m
Top Height, H= 1.9 -m
Slope Angle, As= 0
Batter Angle, Ab= 0

Soil Data:

Depth -m	Gamma -kN/m3	Phi	C -kN/m2	K -MN/m3	e50 or Dr %	Nspt
0	17	27	0.0	6.0	24.26	6
2.25	17	27	0.0	3.9	19.62	5
4.0	7	27	0.0	3.7	19.62	5
4.5	8	34	0.0	23.6	62.40	27
9	8	34	0.0	25.1	64.70	29
13.5	8.5	36	0.0	35.0	77.38	42
18	8.5	36	0.0	42.8	85.61	50

Pile Data:

Depth -m	Width -cm	Area -cm2	Per. -cm	I -cm4	E -MPa	Weight -kN/m
0.0	80	5451.7	251.3	2013680.3	25743	12.394
13.0						

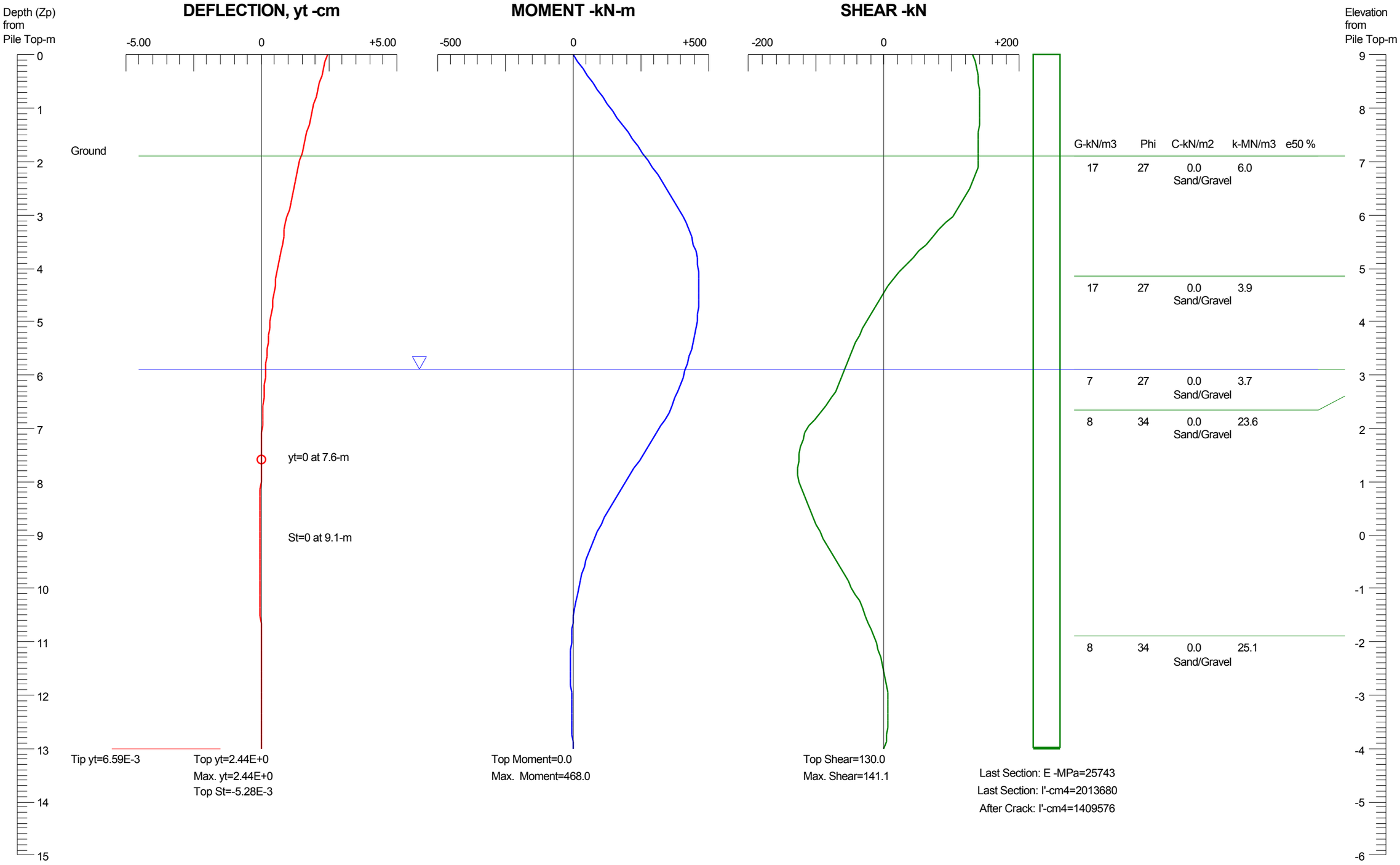
Single Pile Lateral Analysis:

Top Deflection, yt= 2.44000-cm
Max. Moment, M= 468.00-kN-m
Top Deflection Slope, St= -0.00528
OK! Top Deflection, 2.4400-cm is less than the Allowable Deflection= 2.50-cm

Note: If the program cannot find a result or the result exceeds the upper limit. The result will be displayed as 99999.
The Max. Moment calculated by program is an internal force from the applied load conditions. Structural engineer has to check whether the pile has enough capacity to resist the moment with adequate factor of safety. If not, the pile may fail under the load conditions.

PILE DEFLECTION & FORCE vs DEPTH

Single Pile, Khead=1, Kbc=1



CivilTech Software

AMP-3_D800_BH-7
Free End (P=2000kN, V=130kN)

Figure 2

IV) Sample Pile Structural Capacity Check Calculation Sheet

MAADEN PHOSPHATE III PROGRAM

AMMONIA III PROJECT

Title: CALCULATION FOR CAST-IN-PLACE CONCRETE PILE

Number: MD-522-7000-EG-CV-CAL-0001

Date: 14 Feb 2019

Rev: A01

PROPRIETARY & CONFIDENTIAL

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REVISION INDEX

Rev	Date	Description
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DOCUMENT APPRAISAL STAMP

MA'ADEN معادن شركة التعدين العربية السعودية Saudi Arabian Mining Company — JACOBS

REVIEWER'S NAME	SIGN	DATE

REVIEW GRADE

Result	Code	Description
☐	A	Accepted
☐	C	Proceed with exeception to next submission and Revision
☐	R	Do Not Proceed - Revise as noted and Resubmit

Acceptance in any of these categories in no way relieves the Contractor/Supplier of its responsibility for the due & proper performance of the Works in accordance with the Contract/Purchase Order

PROJECT MANAGER SIGN-OFF:

NAME		DATE:

Rev.	Rev. Date	Description	Prepared By	Checked By	Approved By	Employer
A02	27 Feb 2019	Issued for Information	J.H Jeon	M.G Lee	J.J Yang	
A01	14 Feb 2019	Issued for Information	J.H Jeon	M.G Lee	J.J Yang	

CONTENTS

1. DESIGN CRITERIA 3

1.1 DESIGN DATA & MATERIAL PROPERTIES 3
1.2 INPUT LOADS 4
1.3 CALCULATION OF MOMENT IN PILE SHAFT 4

2. DESIGN OF PILE SECTION_D-600 5

2.1 PILE DESIGN FOR CASE 1 5
2.2 PILE DESIGN FOR CASE 2 6
2.3 CALCULATION OF DEVELOPMENT LENGTH 7
2.4 CALCULATION OF SPLICE LENGTH 7
2.5 MEMBER DESIGN 8
2.6 PILE DEFLECTION CHECK 8
2.7 SKETCH 9

3. DESIGN OF PILE SECTION_D-800 10

3.1 PILE DESIGN FOR CASE 1 10
3.2 PILE DESIGN FOR CASE 2 11
3.3 CALCULATION OF DEVELOPMENT LENGTH 12
3.4 CALCULATION OF SPLICE LENGTH 12
3.5 MEMBER DESIGN 13
3.6 PILE DEFLECTION CHECK 13
2.7 SKETCH 14

1. DESIGN CRITERIA

1.1 DESIGN DATA & MATERIAL PROPERTIES

- Type of Pile : Cast-In Place (CIP) Pile

- Diameter & Total Length of Pile : Dp = 600 mm / L_{pile} = 12.0 m
  Dp = 800 mm / L_{pile} = 14.9 m

- Pile Head Elevation : D 600 = + 6.0 m (6.0m above M.S.L, 1.0m below F.G.L)
  D 800 = + 8.9 m (8.9m above M.S.L, 1.9m above F.G.L)

- Compressive Strength of Concrete : fc' = 30 Mpa

- Yield Strength of Longitudinal Reinforcing Steel : fy = 420 Mpa

- Yield Strength of Spiral Reinforcing Steel : fy_{sec} = 420 MPa

- Modulus Elasticity of Reinforcing Steel : Es = 200000 MPa

- Density of Concrete : γc = 24 kN/m^3

- Elastic Modulus of Concrete Ec = 4700 x √(fc') Ec = 25743 MPa

- Provide. Percentage of Longitudinal Reinforcement (considered for vertical comp.member) : P_{t_Pro} = 1.00% ~ 8.00% *(ACI 318M - 14 Cl. 10.6.1.1)*

- Area of Pile Section : Ap = π x Dp^2 / 4
  = 2827.43 & 5026.55 cm^2

- Moment of Inertia of Pile Section : Ip = π x Dp^4 / 64
  = 636172.51 & 2010619.30 cm^4

- Strength reduction Factors *(ACI 318M - 14 Cl. 21.2.1)*

  For Shear : Φ_v = 0.75
  For Bearing : Φ_{bear} = 0.65

1.2 Input Loads

- All loads below are maximum working level loads.

Pile Diameter	Φ 600		Φ 800	
Allowable Static Axial Compression Load, Pc	(1)	1500 kN	(1)	2000 kN
Allowable Static Axial Tensile Load, Pt	(2)	300 kN	(1)	650 kN
Allowable Lateral Load, H	(2)	110 kN	(2)	130 kN

where, (1) : Maximum Load of Geotechnical Report

(2) : The Load to be Utilized for Pile Foundation Design

1.3 Calculation of Moment in Pile Shaft

- Result Summary (from ALL-Pile software analysis, refer to Attachment)

Pile Dia (mm)	Load Type	Head Condition	Moment (kN.m)	Maximum Moment (kN.m)
600	Test	Free	209	209
	Working	Fixed	205	
800	Test	Free	468	468
	Working	Fixed	423	

2. DESIGN OF PILE SECTION_D-600

2.1 PILE DESIGN FOR CASE 1

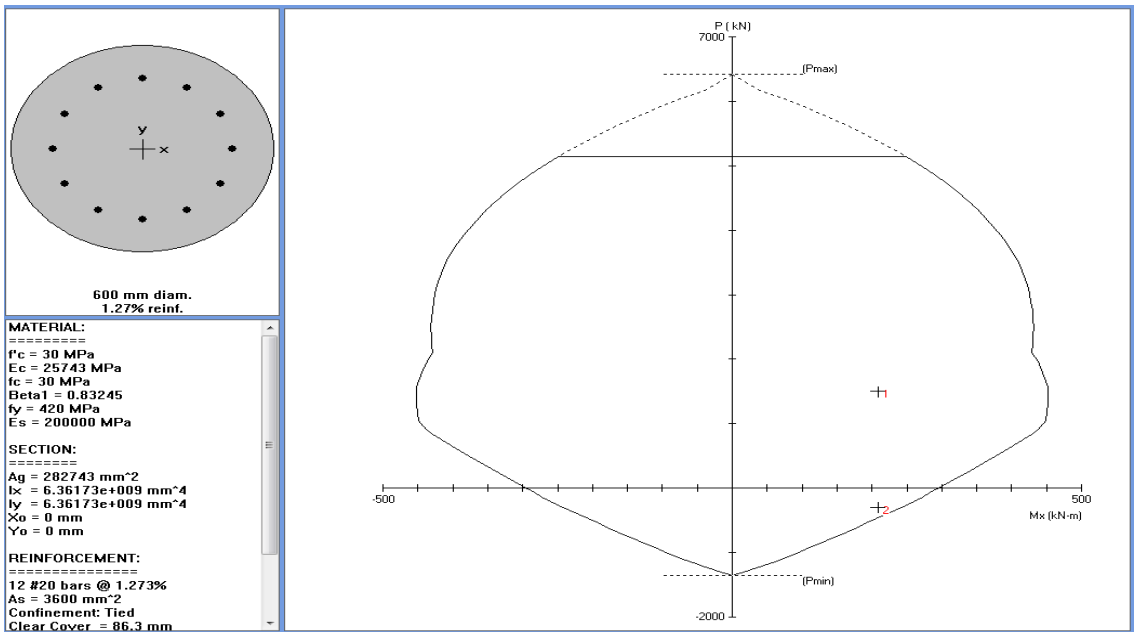
- Design of Pile Reinforcement

Clear cover to rebar cover = 75 mm , Dia of tie bar db1 = 10 mm
Dia of longitudinal rebar Db = 20 mm , No. deformed longitudinal rebars of N = 12 ea

- Area of Longitudinal Reinforcement Use $As = \pi \times Db^2 / 4 \times N$ As = 37.68 cm^2
- Area of Longitudinal Reinforcement Require $Ast = P_{t_Min} \times Ap$ Ast = 28.27 cm^2
- Area of Pile Section Ap = 2827.43 cm^2

- Using. % age of Longitudinal Reinforcement (Considered for Vertical Comp. Member) P_t = 1.33% O.K !
- Provide. % age of Longitudinal Reinforcement (Considered for Vertical Comp. Member) $P_{t_Min} \sim P_{t_Max}$ = 1.00% ~ 8.00%
- Allowable Static Axial Compression Load Pc = 1500.00 kN
- Allowable Static Axial Tensile Load Pt = 300.00 kN
- Bending moment M = 209.00 kN.m

Load Case	Applied Loads	
	Pu (kN)	Mu (kN.m)
1. Allowable Static Axial Compression Load + Bending	1500.00	209.00
2. Allowable Static Axial Tensile Load + Bending	-300.00	209.00



2.2 PILE DESIGN FOR CASE 2

1. Check for Shear with Tension

Vc = H Vc = 110.00 kN Φv = 0.75

Pt = -300.00 kN

- Width of equivalent square section bw = √ Ap bw = 531.74 mm
- Effective depth of equivalent square section

deff = bw - (cover + db + Db x 1/2) deff = 436.74 mm

Vcr = 0.17 x { 1 + (Pt / 3.5 Ap) } x √ f`c x bw x deff Vcr = 150.68 kN

(ACI 318M-14_Cl.22.5.7.1)

Shear capacity of concrete ; Vc = Φv x Vcr Vc = 113.01 kN O.K !

2. Design of Tie Bars

- Tie bars information

Dia of tie db1 = 10 mm

- Dia of Tie bar shall be the largest of the following

a) Vertical Dia Db > 36 mm db1 = 13 mm *(ACI 318M-14_Cl.25.7.2.2)*

b) Vertical Dia Db < 32 mm db1 = 10 mm *(ACI 318M-14_Cl.25.7.2.2)*

- Provide Dia of tie bars as db_{pro} = 10 mm
- Using Dia of tie bars as db_{use} = 10 mm O.K !

- The tie bar are provided in the circular shape
- Spacing of tie bar shall be the least of the following *(ACI 318M-14_Cl.25.7.2.1)*

a) S0 = Dp S0 = 600 mm

b) S1 = 16 x Db S1 = 320 mm

c) S2 = 48 x db1 S2 = 480 mm

- Provide spacing of tie bar as S_{pro} = 320 mm
- Using spacing of tie bar as S_{use} = 300 mm O.K !

2.3 CALCULATION OF DEVELOPMENT LENGTH

1. Material Design Data

(ACI 318M - 14 Cl. 25.4.2.4)

- Compressive strength of concrete — $f'c$ = 30 MPa
- Yield strength of reinforcement — fy = 420 MPa
- Nominal daimeter of used bar — Db = 20 mm → ψ_s = 1.00
- Center-to-center spacing of bars — s = 106.1 mm
- Cover depth — $cover$ = 75 mm
- Rebar Location Factor — Ψt = 1.00 (Other bar)
- Rebar Coating Factor — Ψe = 1.50 (For epoxy-coated bars or clear spacing less than 6db)
- Aggregate Weight Factor — λ = 1.00 (Normal weight)

2. Design of Development Length for Tension & hooks

- C_b = Min.[cover + D_b/2 or s/2] — C_b = Min.[85mm or 53mm] = 53.06 mm
- Transverse reinforcement index — K_{tr} = 0 (It shall be permitted to use K_{tr}= 0 as a design simplification)
  *α = (C_b + K_{tr}) / D_b = (53 + 0) / 20 = 2.65 (should be ≤ 2.50) → Use α = 2.50
- Development Length for Compression

  Ld_{comp} = Max [(0.24 x fy / λ x √f'c) x Db , 0.043 x fy x Db , 200]

  Development Length for Required Ld_{comp} = 370 mm *(ACI 318M - 14 Cl. 25.4.9)*
- Development Length for Tension

  Ld_{ten} = Max [(fy x ψt x ψe x ψs / 1.1 x λ x √f'c x α) x Db , 300] *(ACI 318M - 14 Cl. 25.4.2)*

  Ld_{ten} = 840 mm
- The actual stress in the reinforcement bar shall be lesser than fy due to the larger section and more reinforcement provided than actually required from the strength calculations.

  Hence development length in tension for the reinforcement bar at actual stress level

  Ld_{ten_act} = Ld_{ten} x Ast / As — Ld_{ten_act} = 630 mm *(ACI 318M - 14 Cl. 25.4.10)*
- Provided Straight length of bar above Cut Off Level — $L_{straight}$ = 650 mm O.K !

- Development of standard hooks in tension

  L_{dh} = Max [(0.24 x fy x ψe / λ x √f'c x Db , 8 Db , 150] *(ACI 318M - 14 Cl. 25.4.3.1)*

  ψe : 1.20 (epoxy-coated)

  AF = 0.70 ; normal to plane of hook (65mm) ≤ 75 mm

  90 degree hook with cover on bar (50mm) ≤ 75 mm

  L_{bent} = L_{dh} x AF (Applicable factors)

  bar length bent 90 deg. at top — L_{bent} = 310 mm *(ACI 318M - 14 Cl. 25.4.3)*

2.4 CALCULATION OF SPLICE LENGTH

- "Class B" lap splice is provided for the main rebars as per the requirements *(ACI 318M - 14 Cl. 25.5.2.1)*

  Length_splice = 1.3 x Ld_{ten} — $Length_{splice}$ = 1100 mm

  Provided length of splice — $Length_{splice\ prov}$ = 1100 mm O.K !

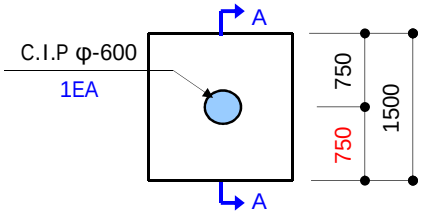
2.5 MEMBER DESIGN

- Diameter of Pile (Dp) = 600 mm
- Total length of Pile considered (Lpile) = 12 m
- Vertical Re-bar (Db) = 12 ea - D 20
- Ties Re-bar = D 10 @ 300
- Splice Length (Length_splice prov) = 1100 mm (Vertical Rebar)
- Clear cover to rebar = 75 mm

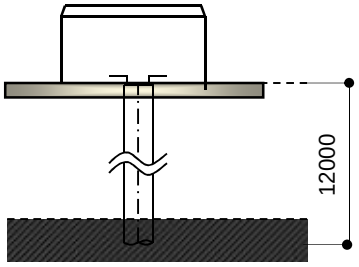
2.6 PILE DEFLECTION CHECK

1. ALL-Pile

[Program Analysis Plan] [unit : mm]



Plan & Pile Arrangement



Section A-A

[Program Output] *(Refer to. Appendix)*

Soil Data:

Depth -m	Gamma -kN/m3	Phi	C -kN/m2	K -MN/m3	e50 or Dr %	Nspt
0	17	27	0.0	4.5	20.99	5
2.25	18	34	0.0	22.7	47.56	16
4.0	8	34	0.0	14.8	47.73	16
4.5	8	34	0.0	15.6	49.28	17
9	8.5	36	0.0	27.3	67.76	32
13.5	8.5	36	0.0	36.5	79.08	44
18	8.5	36.0	0.0	42.8	85.83	50

Pile Data:

Depth -m	Width -cm	Area -cm2	Per. -cm	I -cm4	E -MPa	Weight -kN/m
0.0	60	3082.5	188.5	637274.4	25743	6.984
12.0						

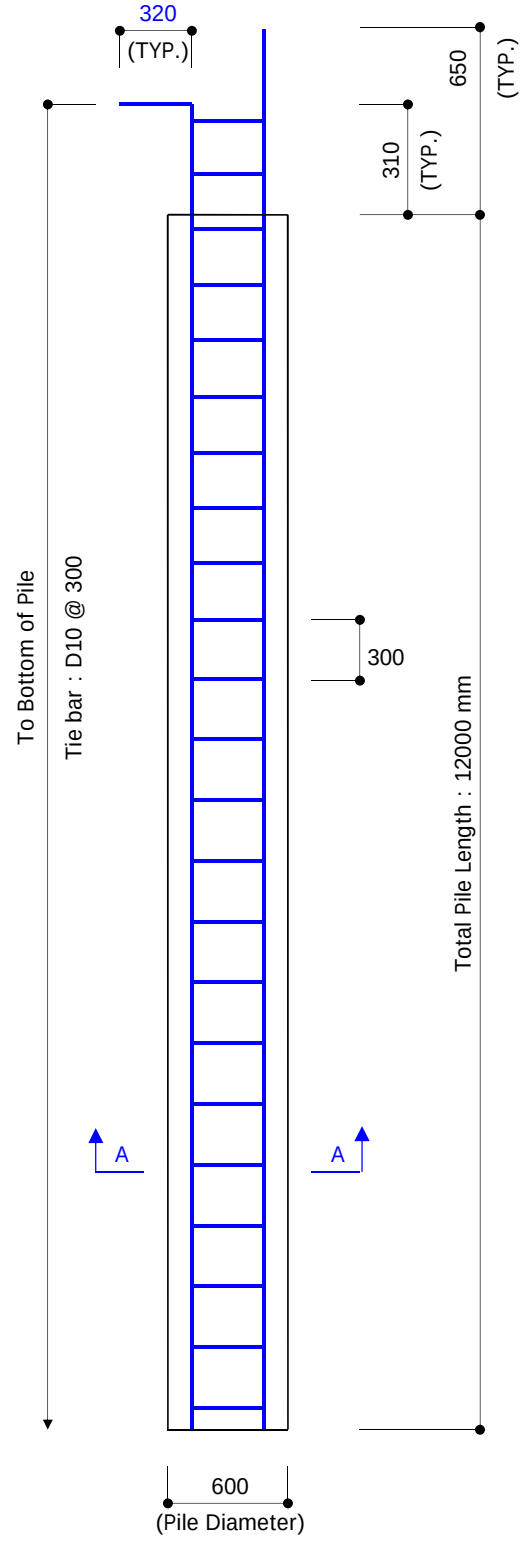
Single Pile Lateral Analysis:
Top Deflection, yt= 1.21000-cm
Max. Moment, M= 209.00-kN-m
Top Deflection Slope, St= -0.00445
OK! Top Deflection, 1.2100-cm is less than the Allowable Deflection= 2.50-cm

2. Result Summary

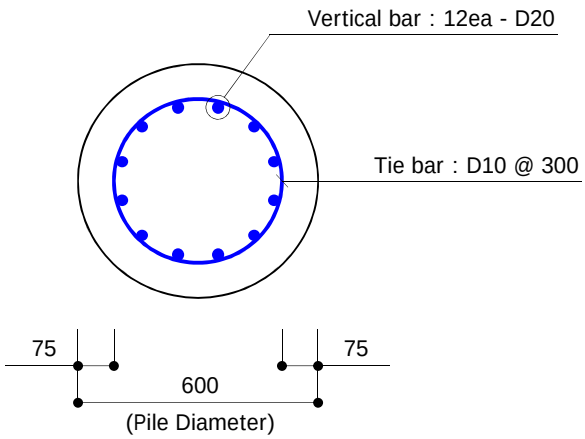
Pile-head Deflection ALL-Pile (mm)	Allowable Deflection (mm)	Check
12.00	25.00	O.K !

*Allowable Deflection : 25 mm

2.7 SKETCH



- LONGITUDINAL SECTION -



- SECTION A -

3. DESIGN OF PILE SECTION_D-800

3.1 PILE DESIGN FOR CASE 1

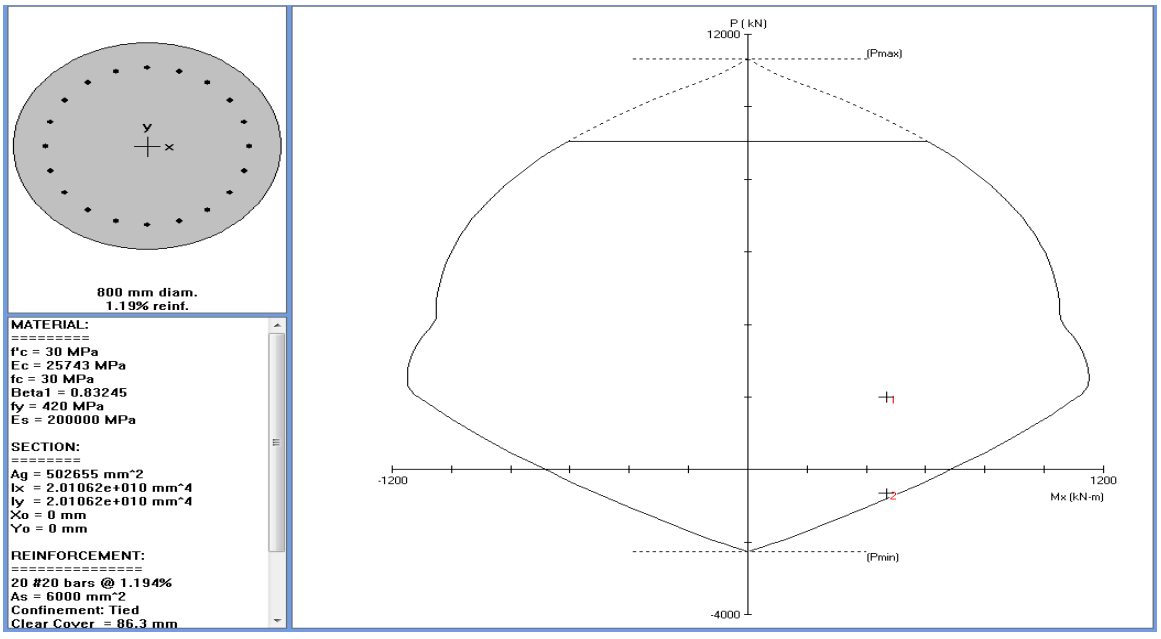
- Design of Pile Reinforcement

Clear cover to rebar cover = 75 mm , Dia of tie bar db1 = 10 mm
Dia of longitudinal rebar Db = 20 mm , No. deformed longitudinal rebars of N = 20 ea

- Area of Longitudinal Reinforcement Use As = π x Db² / 4 x N As = 62.80 cm^2
- Area of Longitudinal Reinforcement Require Ast = P_{t_Min} x Ap Ast = 50.27 cm^2
- Area of Pile Section Ap = 5026.55 cm^2

- Using. % age of Longitudinal Reinforcement (Considered for Vertical Comp. Member) P_t = 1.25% O.K !
- Provide. % age of Longitudinal Reinforcement (Considered for Vertical Comp. Member) P_{t_Min} ~ P_{t_Max} = 1.00% ~ 8.00%
- Allowable Static Axial Compression Load Pc = 2000.00 kN
- Allowable Static Axial Tensile Load Pt = 650.00 kN
- Bending moment M = 468.00 kN.m

Load Case	Applied Loads	
	Pu (kN)	Mu (kN.m)
1. Allowable Static Axial Compression Load + Bending	2000.00	468.00
2. Allowable Static Axial Tensile Load + Bending	-650.00	468.00



3.2 PILE DESIGN FOR CASE 2

1. Check for Shear with Tension

Vc = H Vc = 130.00 kN Φv = 0.75

Pt = -650.00 kN

- Width of equivalent square section bw = √ Ap bw = 708.98 mm
- Effective depth of equivalent square section

deff = bw - (cover + db + Db x 1/2) deff = 613.98 mm

Vcr = 0.17 x { 1 + (Pt / 3.5 Ap) } x √ f`c x bw x deff Vcr = 255.57 kN

(ACI 318M-14_Cl.22.5.7.1)

Shear capacity of concrete ; Vc = Φv x Vcr Vc = 191.68 kN O.K !

2. Design of Tie Bars

- Tie bars information

Dia of tie db1 = 10 mm

- Dia of Tie bar shall be the largest of the following

a) Vertical Dia Db > 32 mm db1 = 12 mm *(ACI 318M-14_Cl.25.7.2.2)*

b) Vertical Dia Db < 32 mm db1 = 10 mm *(ACI 318M-14_Cl.25.7.2.2)*

- Provide Dia of tie bars as db_{pro} = 10 mm
- Using Dia of tie bars as db_{use} = 10 mm O.K !

- The tie bar are provided in the circular shape
- Spacing of tie bar shall be the least of the following *(ACI 318M-14_Cl.25.7.2.1)*

a) S0 = Dp S0 = 800 mm

b) S1 = 16 x Db S1 = 320 mm

c) S2 = 48 x db1 S2 = 480 mm

- Provide spacing of tie bar as S_{pro} = 320 mm
- Using spacing of tie bar as S_{use} = 300 mm O.K !

3.3 CALCULATION OF DEVELOPMENT LENGTH

1. Material Design Data

- Compressive strength of concrete $f'c = 30$ MPa
- Yield strength of reinforcement $fy = 420$ MPa
- Nominal daimeter of used bar $Db = 20$ mm → $\psi_s = 1.00$
- Center-to-center spacing of bars $s = 95.43$ mm
- Cover depth $cover = 75$ mm
- Rebar Location Factor $\Psi t = 1.00$ (Other bar)
- Rebar Coating Factor $\Psi e = 1.50$ (For epoxy-coated bars or clear spacing less than 6db)

- Aggregate Weight Factor $\lambda = 1.00$ (Normal weight)

2. Design of Development Length for Tension & hooks

- C_b = Min.[cover + D_b/2 or s/2] C_b = Min.[85mm or 48mm] = 47.71 mm
- Transverse reinforcement index $K_{tr} = 0$ (It shall be permitted to use K_{tr}= 0 as a design simplification)
  *$\alpha = (C_b + K_{tr}) / D_b = (48 + 0) / 20 = 2.39$ (should be ≤ 2.50) → Use $\alpha = 2.39$
- Development Length for Compression
  Ld_{comp} = Max [(0.24 x fy / λ x √f'c) x Db , 0.043 x fy x Db , 200]
  Development Length for Required Ld_{comp} = 370 mm *(ACI 318M - 14 Cl. 25.4.9)*
- Development Length for Tension
  Ld_{ten} = Max [(fy x ψt x ψe x ψs / 1.1 x λ x √f'c x α) x Db , 300] *(ACI 318M - 14 Cl. 25.4.2)*
  Ld_{ten} = 880 mm
- The actual stress in the reinforcement bar shall be lesser than fy due to the larger section and more reinforcement provided than actually required from the strength calculations.
  Hence development length in tension for the reinforcement bar at actual stress level
  $Ld_{ten_act} = Ld_{ten}$ x Ast / As Ld_{ten_act} = 704 mm *(ACI 318M - 14 Cl. 25.4.10)*
- Provided Straight length of bar above Cut Off Level $L_{straight}$ = 710 mm O.K !

- Development of standard hooks in tension
  L_{dh} = Max [(0.24 x fy x ψe / λ x √f'c x Db , 8 Db , 150] *(ACI 318M - 14 Cl. 25.4.3.1)*
  ψe : 1.20 (epoxy-coated)
  AF = 0.70 ; normal to plane of hook (65mm) ≤ 75 mm
  90 degree hook with cover on bar (50mm) ≤ 75 mm
  $L_{bent} = L_{dh}$ x AF (Applicable factors)
  bar length bent 90 deg. at top L_{bent} = 310 mm *(ACI 318M - 14 Cl. 25.4.3)*

3.4 CALCULATION OF SPLICE LENGTH

- "Class B" lap splice is provided for the main rebars as per the requirements *(ACI 318M - 14 Cl. 25.5.2.1)*
  Length_splice = 1.3 x Ld_{ten} $Length_{splice}$ = 1150 mm
  Provided length of splice $Length_{splice\ prov}$ = 1200 mm O.K !

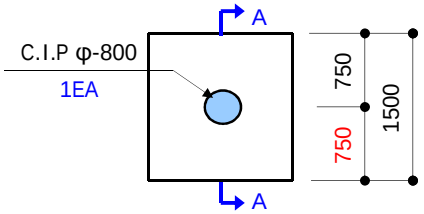
3.5 MEMBER DESIGN

- Diameter of Pile (Dp) = 800 mm
- Total length of Pile considered (Lpile) = 14.9 m
- Vertical Re-bar (db) = 20 ea - D 20
- Ties Re-bar = D 10 @ 300
- Splice Length (Length_splice prov) = 1200 mm (Vertical Rebar)
- Clear cover to rebar = 75 mm

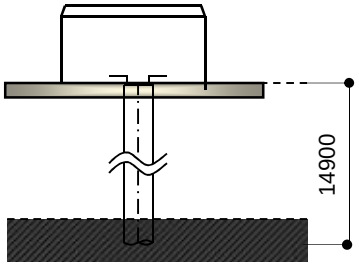
3.6 PILE DEFLECTION CHECK

1. ALL-Pile

[Program Analysis Plan] [unit : mm]



Plan & Pile Arrangement



Section A-A

[Program Output] *(Refer to. Appendix)*

Soil Data:

Depth -m	Gamma -kN/m3	Phi	C -kN/m2	K -MN/m3	e50 or Dr %	Nspt
0	17	27	0.0	6.0	24.26	6
2.25	17	27	0.0	3.9	19.62	5
4.0	7	27	0.0	3.7	19.62	5
4.5	8	34	0.0	23.6	62.40	27
9	8	34	0.0	25.1	64.70	29
13.5	8.5	36	0.0	35.0	77.38	42
18	8.5	36	0.0	42.8	85.61	50

Pile Data:

Depth -m	Width -cm	Area -cm2	Per. -cm	I -cm4	E -MPa	Weight -kN/m
0.0	80	5451.7	251.3	2013680.3	25743	12.394
13.0						

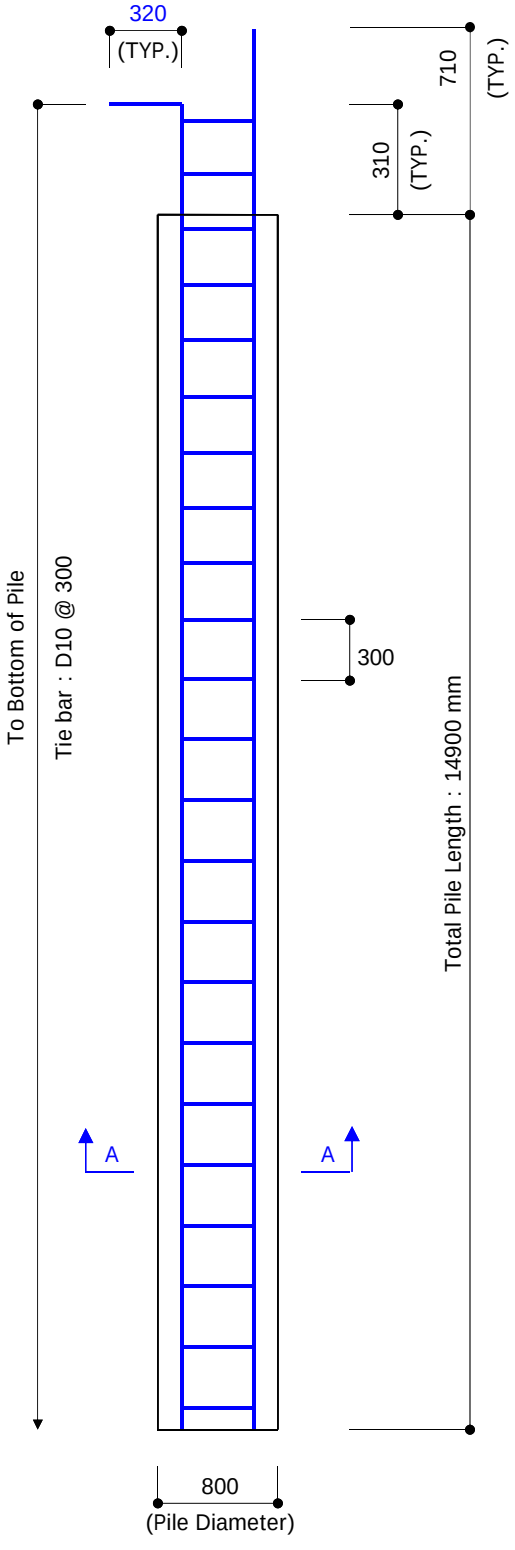
Single Pile Lateral Analysis:
Top Deflection, yt= 2.44000-cm
Max. Moment, M= 468.00-kN-m
Top Deflection Slope, St= -0.00528
OK! Top Deflection, 2.4400-cm is less than the Allowable Deflection= 2.50-cm

2. Result Summary

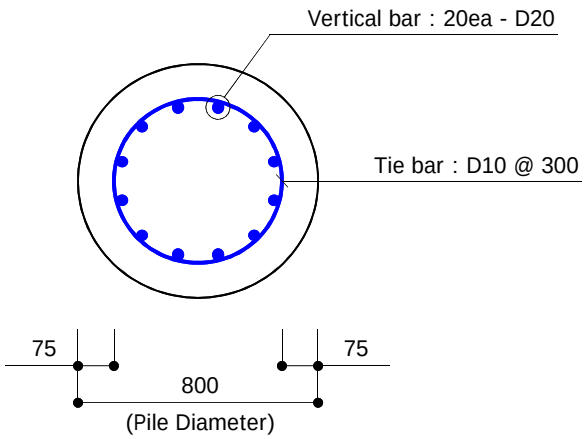
Pile-head Deflection ALL-Pile (mm)	Allowable Deflection (mm)	Check
24.40	25.00	O.K !

*Allowable Deflection : 25 mm

2.7 SKETCH



- LONGITUDINAL SECTION -



- SECTION A -

Attachment-1: Boring Hole Log

Attachment-2: Laboratory Test Result

ATTACHMENT-3 : PILE CAPACITY ANALYSIS RESULTS FOR ALL BH LOGS EXCEPT BH ON FUTURE AREA